Complex Analysis

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Introduction

Complex analysis studies functions

$$f : \Omega \subset \mathbb{C} \longrightarrow \mathbb{C}$$

that are differentiable with respect to a complex variable. At first sight, this appears to be only a two-dimensional version of real calculus. The definition

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

even looks identical to the usual definition of a derivative. The essential difference is that $h \in \mathbb{C}$ may approach 0 from infinitely many directions. Requiring all of these approaches to give the same limit is an extremely strong condition.

This rigidity is the starting point of complex analysis. Writing

$$f(z) = u(x, y) + iv(x, y), \qquad z = x + iy,$$

complex differentiability forces the real and imaginary parts to satisfy the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Thus a local condition on a difference quotient immediately imposes a system of partial differential equations. More remarkably, once a function is holomorphic, it is not merely once differentiable: it is infinitely complex differentiable and locally represented by a convergent power series,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

In this sense, holomorphic functions are far more rigid than ordinary smooth functions of real variables.

The central mechanism behind this rigidity is integration in the complex plane. For a parametrized curve γ, one defines

$$\int_\gamma f(z)\ \mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t.$$

The fundamental phenomenon is that, under appropriate hypotheses, holomorphic functions have vanishing integrals around closed curves:

$$\int_\gamma f(z)\ \mathrm{d}z = 0.$$

Cauchy's theorem turns this apparently simple statement into one of the main structural principles of the subject. Its most important consequence is Cauchy's integral formula,

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0}\ \mathrm{d}z,$$

which says that the value of a holomorphic function at an interior point is completely determined by its values along a surrounding curve.

This formula illustrates one of the most striking features of complex analysis: *boundary information controls interior behavior*. By differentiating the formula one obtains all

derivatives of f; from these follow Cauchy's inequalities, Liouville's theorem, the fundamental theorem of algebra, and uniqueness principles for holomorphic functions. A function defined on a two-dimensional region can therefore be constrained by its behavior on a one-dimensional boundary.

Singularities reveal an equally important side of the theory. A function may fail to be holomorphic at isolated points, but such failures are highly structured. Near an isolated singularity z_0, a Laurent expansion takes the form

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

The coefficient

$$a_{-1} = \operatorname{Res}_{z=z_0} f$$

is the *residue* of f at z_0. Although it is only one coefficient among infinitely many, it contains precisely the information needed to compute the contribution of the singularity to a contour integral.

This culminates in the residue theorem,

$$\boxed{\int_\gamma f(z)\, \mathrm{d}z = 2\pi i \sum_{z_k \text{ inside } \gamma} \operatorname{Res}_{z=z_k} f}.$$

The beauty of this theorem is the conversion of a global problem into local data. Instead of evaluating an integral along an entire curve, one identifies the isolated singularities inside the curve and computes a single coefficient at each of them. Complicated real integrals can consequently be transformed into algebraic calculations involving poles and residues. This local-to-global principle is one of the recurring themes of complex analysis.

The same philosophy appears in the argument principle: zeros and poles inside a curve can be counted by the boundary integral

$$\frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)}\, \mathrm{d}z.$$

Thus integration detects not only values of a function, but also geometric and topological information about its zeros and singularities.

Finally, complex analysis has a geometric interpretation through conformal mappings. A holomorphic function with nonzero derivative preserves angles locally, allowing complicated planar domains to be transformed into simpler ones such as the unit disc

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

or the upper half-plane

$$\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}.$$

Möbius transformations, the Schwarz lemma, automorphisms of $\mathbb{D}$ and $\mathbb{H}$, and the Riemann mapping theorem show that analytic and geometric structures in the complex plane are deeply intertwined.

The main ideas of an undergraduate course in complex analysis can therefore be summarized by the chain

$$\boxed{\begin{array}{c} \text{complex differentiability} \Longrightarrow \text{holomorphicity and analyticity} \\ \Longrightarrow \text{Cauchy theory} \Longrightarrow \text{strong rigidity} \\ \Longrightarrow \text{zeros and singularities} \Longrightarrow \text{residues} \Longrightarrow \text{global information}, \end{array}}$$

together with the geometric viewpoint provided by conformal mappings. The purpose of these notes is to develop these connections while emphasizing the definitions, fundamental theorems, representative examples, and techniques that form the core of undergraduate complex analysis.

Chapter 1

Complex Numbers and Holomorphic Functions

1 Complex Numbers and Holomorphic Functions

1.1 Preliminaries on the Complex Plane

We begin with the basic algebraic and topological structure of the complex plane. These notions will be used throughout the study of holomorphic functions.

Definition 1.1 (Complex Numbers)**.** A complex number is written as

$$z = x + iy, \qquad x, y \in \mathbb{R},$$

where $i^2 = -1$. We define

$$\Re(z) = x, \qquad \Im(z) = y.$$

The real axis consists of the complex numbers with $\Im(z) = 0$, while the imaginary axis consists of those with $\Re(z) = 0$.

Addition and multiplication of complex numbers satisfy the usual commutative, associative, and distributive laws.

Definition 1.2 (Modulus)**.** For $z = x + iy \in \mathbb{C}$, its modulus is

$$|z| = (x^2 + y^2)^{1/2}.$$

Geometrically, $|z|$ is the Euclidean distance from the point z to the origin in the complex plane.

The modulus satisfies the triangle inequalities

$$|z + w| \le |z| + |w|$$

and

$$\big||z| - |w|\big| \le |z - w|.$$

Definition 1.3 (Complex Conjugate)**.** For

$$z = x + iy,$$

the complex conjugate of z is

$$\overline{z} = x - iy.$$

The conjugate satisfies

$$\overline{\overline{z}} = z, \qquad |z|^2 = z\overline{z},$$

and

$$\Re(z) = \frac{z + \overline{z}}{2}, \qquad \Im(z) = \frac{z - \overline{z}}{2i}.$$

Remark 1.4. Complex conjugation is geometrically reflection across the real axis. Consequently, $|z| = |\overline{z}|$.

Definition 1.5 (Polar Form)**.** Every nonzero complex number can be written as

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta),$$

where

$$r = |z| > 0.$$

The number θ is called an argument of z.

The representation of θ is not unique, since

$$e^{i(\theta+2\pi k)} = e^{i\theta}, \qquad k \in \mathbb{Z}.$$

This non-uniqueness will later be responsible for the multivalued nature of the complex logarithm.

1.2 Convergence and Basic Topology

Definition 1.6 (Convergence)**.** A sequence $\{z_n\} \subset \mathbb{C}$ converges to $w \in \mathbb{C}$ if

$$\lim_{n\to\infty} |z_n - w| = 0.$$

We write

$$z_n \to w \qquad \text{or} \qquad w = \lim_{n\to\infty} z_n.$$

Since

$$|z_n - w|^2 = |\Re(z_n) - \Re(w)|^2 + |\Im(z_n) - \Im(w)|^2,$$

complex convergence is equivalent to convergence of the real and imaginary parts separately.

Proposition 1.7. *Let*

$$z_n = x_n + iy_n, \qquad z = x + iy.$$

Then

$$z_n \to z$$

if and only if

$$x_n \to x \qquad \textit{and} \qquad y_n \to y.$$

Definition 1.8 (Cauchy Sequence)**.** A sequence $\{z_n\} \subset \mathbb{C}$ is a Cauchy sequence if

$$|z_n - z_m| \longrightarrow 0 \qquad \text{as } n, m \to \infty.$$

Theorem 1.9 (Completeness of $\mathbb{C}$)**.** *Every Cauchy sequence in* $\mathbb{C}$ *converges to a point in* $\mathbb{C}$*.*

Remark 1.10. This follows from the completeness of $\mathbb{R}$, since the real and imaginary parts of a Cauchy sequence in $\mathbb{C}$ are Cauchy sequences in $\mathbb{R}$.

Definition 1.11 (Open and Closed Discs)**.** For $z_0 \in \mathbb{C}$ and $r > 0$, the open disc centered at z_0 with radius r is

$$D_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\}.$$

The closed disc is

$$\overline{D}_r(z_0) = \{z \in \mathbb{C} : |z - z_0| \le r\},$$

and its boundary circle is

$$C_r(z_0) = \{z \in \mathbb{C} : |z - z_0| = r\}.$$

The unit disc is

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

Definition 1.12 (Interior Point)**.** A point $z_0 \in \Omega$ is an interior point of Ω if there exists $r > 0$ such that

$$D_r(z_0) \subset \Omega.$$

Definition 1.13 (Open Set)**.** A set $\Omega \subset \mathbb{C}$ is open if every point of Ω is an interior point.

Definition 1.14 (Limit Point)**.** A point $z_0 \in \mathbb{C}$ is a limit point of $E \subset \mathbb{C}$ if every disc $D_r(z_0)$ contains a point of E different from z_0.

Definition 1.15 (Closed Set)**.** A set $E \subset \mathbb{C}$ is closed if it contains all of its limit points.

Definition 1.16 (Boundary)**.** The boundary of Ω, denoted by $\partial\Omega$, consists of the points $z \in \mathbb{C}$ such that every disc centered at z intersects both Ω and its complement.

Definition 1.17 (Bounded Set)**.** A set $\Omega \subset \mathbb{C}$ is bounded if there exists $M > 0$ such that

$$|z| < M$$

for every $z \in \Omega$.

Definition 1.18 (Diameter)**.** The diameter of a bounded set $E \subset \mathbb{C}$ is

$$\operatorname{diam}(E) = \sup_{z,w \in E} |z - w|.$$

Definition 1.19 (Compact Set)**.** A set $K \subset \mathbb{C}$ is compact if it is closed and bounded.

Theorem 1.20 (Characterizations of Compactness)**.** *For a set $K \subset \mathbb{C}$, the following properties are equivalent:*

1. *K is compact;*
2. *every sequence in K has a subsequence converging to a point of K;*
3. *every open covering of K has a finite subcovering.*

Remark 1.21. In $\mathbb{C} \simeq \mathbb{R}^2$, the statement that compact sets are precisely the closed and bounded sets is the Heine–Borel theorem.

Theorem 1.22 (Nested Compact Sets)**.** *Let*

$$K_1 \supset K_2 \supset K_3 \supset \cdots$$

be nonempty compact subsets of $\mathbb{C}$. If

$$\operatorname{diam}(K_n) \longrightarrow 0,$$

then there exists a unique $w \in \mathbb{C}$ such that

$$w \in K_n$$

for every n.

Proof. Choose $z_n \in K_n$. Since K_1 is compact, the sequence $\{z_n\} \subset K_1$ has a convergent subsequence

$$z_{n_k} \to w \in K_1.$$

For any fixed m, we have $z_{n_k} \in K_m$ whenever $n_k \geq m$. Since K_m is closed, $w \in K_m$. Hence

$$w \in \bigcap_{m=1}^{\infty} K_m.$$

If both w_1 and w_2 belong to every K_n, then

$$|w_1 - w_2| \leq \operatorname{diam}(K_n)$$

for every n. Letting $n \to \infty$ gives

$$|w_1 - w_2| = 0.$$

Therefore $w_1 = w_2$. □

Definition 1.23 (Connected Set)**.** A set $\Omega \subset \mathbb{C}$ is connected if it is not possible to find two disjoint nonempty open sets Ω_1, Ω_2 such that

$$\Omega = \Omega_1 \cup \Omega_2.$$

Definition 1.24 (Region)**.** A region is a connected open subset of $\mathbb{C}$.

Remark 1.25. The assumptions "open" and "connected" play different roles. Openness allows local analytic arguments using small discs, while connectedness allows local information to propagate through the entire domain. This distinction becomes important in results such as the identity theorem.

1.3 Continuity of Complex Functions

Definition 1.26 (Continuity)**.** Let $f : \Omega \to \mathbb{C}$, and let $z_0 \in \Omega$. The function f is continuous at z_0 if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \quad \Longrightarrow \quad |f(z) - f(z_0)| < \varepsilon.$$

Equivalently,

$$\lim_{z \to z_0} f(z) = f(z_0).$$

Writing

$$f(z) = u(x, y) + iv(x, y),$$

continuity of f is equivalent to continuity of both u and v.

1.4 Complex Differentiability and Holomorphic Functions

The definition of the complex derivative looks identical to the derivative from real calculus, but it is much more restrictive because the increment h may approach 0 through any direction in the complex plane.

Definition 1.27 (Complex Derivative)**.** Let $f : \Omega \to \mathbb{C}$ and let $z_0 \in \Omega$ be an interior point. We define

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h},$$

provided that the limit exists.

Definition 1.28 (Holomorphic Function)**.** A function f is holomorphic in an open set Ω if it is complex differentiable at every point of Ω.

Definition 1.29 (Entire Function)**.** A function f is entire if it is holomorphic in all of $\mathbb{C}$.

Example 1.30. The function

$$f(z) = z^2$$

is entire. Indeed,

$$\frac{f(z + h) - f(z)}{h} = \frac{(z + h)^2 - z^2}{h} = 2z + h,$$

so

$$f'(z) = 2z.$$

Example 1.31. The function

$$f(z) = \frac{1}{z}$$

is holomorphic on

$$\mathbb{C} \setminus \{0\},$$

since

$$\begin{aligned} f'(z) &= \lim_{h \to 0} \frac{\frac{1}{z+h} - \frac{1}{z}}{h} \\ &= \lim_{h \to 0} \frac{-1}{z(z+h)} = -\frac{1}{z^2}. \end{aligned}$$

Example 1.32 (A Continuous Function That Is Not Holomorphic)**.** Consider

$$f(z) = \overline{z}.$$

At z_0,

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\overline{h}}{h}.$$

If $h \in \mathbb{R}$, then

$$\frac{\overline{h}}{h} = 1.$$

If $h = it$, $t \in \mathbb{R}$, then

$$\frac{\overline{h}}{h} = \frac{-it}{it} = -1.$$

Hence the limit depends on the direction in which $h \to 0$. Therefore

$$f(z) = \overline{z}$$

is nowhere holomorphic.

Remark 1.33. This example illustrates the essential difference between real and complex differentiability. Along each individual line one may obtain a directional derivative, but complex differentiability requires all possible directions to give exactly the same complex derivative.

Proposition 1.34 (Algebra of Holomorphic Functions)**.** *If f and g are holomorphic in Ω, then*

$$f + g, \qquad f - g, \qquad fg$$

are holomorphic in Ω. If $g(z) \neq 0$, then

$$\frac{f}{g}$$

is holomorphic at z.

Their derivatives satisfy

$$\begin{aligned} (f+g)' &= f' + g', \\ (fg)' &= f'g + fg', \end{aligned}$$

and

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Proposition 1.35 (Chain Rule)**.** *Let*

$$f : \Omega \to U, \qquad g : U \to \mathbb{C}$$

be holomorphic. Then $g \circ f$ is holomorphic and

$$(g \circ f)'(z) = g'(f(z))f'(z).$$

1.5 Real Differentiability and the Cauchy–Riemann Equations

Let

$$z = x + iy$$

and write

$$f(z) = u(x, y) + iv(x, y).$$

Thus a complex-valued function may be regarded as a map

$$F : \mathbb{R}^2 \to \mathbb{R}^2, \qquad F(x, y) = \begin{pmatrix} u(x, y) \\ v(x, y) \end{pmatrix}.$$

Definition 1.36 (Real Differentiability)**.** The map $F : \mathbb{R}^2 \to \mathbb{R}^2$ is differentiable at p_0 if there exists a linear transformation J such that

$$F(p_0 + h) - F(p_0) - J(h) = o(|h|)$$

as $h \to 0$.

The corresponding Jacobian matrix is

$$J_F(x, y) = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

Theorem 1.37 (Cauchy–Riemann Equations)**.** *Suppose that*

$$f(z) = u(x, y) + iv(x, y)$$

is holomorphic at $z_0 = x_0 + iy_0$. Then

$$\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0),$$

and

$$\frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0).$$

Moreover,

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0)$$

and equivalently

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

Proof. Approach z_0 first through real increments $h \in \mathbb{R}$. Then

$$\begin{aligned} f'(z_0) &= \lim_{h\to 0} \frac{u(x_0+h, y_0) - u(x_0, y_0) + i[v(x_0+h, y_0) - v(x_0, y_0)]}{h} \\ &= u_x(x_0, y_0) + iv_x(x_0, y_0). \end{aligned}$$

Now approach z_0 through purely imaginary increments ih, where $h \in \mathbb{R}$. Then

$$\begin{aligned} f'(z_0) &= \lim_{h\to 0} \frac{u(x_0, y_0+h) - u(x_0, y_0) + i[v(x_0, y_0+h) - v(x_0, y_0)]}{ih} \\ &= v_y(x_0, y_0) - iu_y(x_0, y_0). \end{aligned}$$

Since the complex derivative must be independent of the direction of approach,

$$u_x + iv_x = v_y - iu_y.$$

Comparing real and imaginary parts gives

$$u_x = v_y, \qquad u_y = -v_x.$$

□

The converse requires slightly more than merely knowing that the four partial derivatives exist.

Theorem 1.38 (Sufficient Condition for Holomorphicity)**.** *Suppose*

$$f(z) = u(x, y) + iv(x, y),$$

where the first partial derivatives of u and v are continuous in a neighborhood of z_0. If the Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

hold at z_0, then f is complex differentiable at z_0.

Proof. Let

$$h = \Delta x + i\Delta y.$$

Real differentiability gives

$$\begin{aligned} \Delta u &= u_x\Delta x + u_y\Delta y + o(|h|), \\ \Delta v &= v_x\Delta x + v_y\Delta y + o(|h|). \end{aligned}$$

Therefore

$$f(z_0+h) - f(z_0) = (u_x + iv_x)\Delta x + (u_y + iv_y)\Delta y + o(|h|).$$

Using

$$v_x = -u_y, \qquad v_y = u_x,$$

we obtain

$$\begin{aligned} f(z_0+h) - f(z_0) &= (u_x + iv_x)(\Delta x + i\Delta y) + o(|h|) \\ &= (u_x + iv_x)h + o(|h|). \end{aligned}$$

Hence

$$\frac{f(z_0+h) - f(z_0)}{h} = u_x + iv_x + \frac{o(|h|)}{h}.$$

Since

$$\left|\frac{o(|h|)}{h}\right| \longrightarrow 0,$$

we conclude that

$$f'(z_0) = u_x + iv_x.$$

□

Remark 1.39. The Cauchy–Riemann equations alone, without an appropriate regularity assumption, do not automatically imply complex differentiability.

Thus one should distinguish carefully between

$$\text{holomorphic} \Longrightarrow \text{Cauchy–Riemann equations}$$

and the converse statement, which normally requires additional assumptions such as continuity of the first partial derivatives.

Example 1.40. Consider

$$f(z) = z^2.$$

Writing

$$z = x + iy,$$

we obtain

$$f(z) = x^2 - y^2 + 2ixy.$$

Hence

$$u(x, y) = x^2 - y^2, \qquad v(x, y) = 2xy.$$

Then

$$u_x = 2x = v_y,$$

and

$$u_y = -2y = -v_x.$$

Therefore the Cauchy–Riemann equations hold everywhere.

Example 1.41. For

$$f(z) = \overline{z} = x - iy,$$

we have

$$u(x, y) = x, \qquad v(x, y) = -y.$$

Therefore

$$u_x = 1, \qquad v_y = -1,$$

so the first Cauchy–Riemann equation fails everywhere. This gives another proof that $z \mapsto \overline{z}$ is nowhere holomorphic.

1.6 Wirtinger Derivatives

It is often convenient to express the Cauchy–Riemann equations using the operators

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$$

and

$$\frac{\partial}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

Proposition 1.42. *Let*

$$f = u + iv$$

be continuously differentiable. Then

$$f \text{ is holomorphic} \quad \Longleftrightarrow \quad \frac{\partial f}{\partial \overline{z}} = 0.$$

When f is holomorphic,

$$f'(z) = \frac{\partial f}{\partial z}.$$

Proof. We compute

$$\begin{aligned}\frac{\partial f}{\partial \overline{z}} &= \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)(u + iv) \\ &= \frac{1}{2}\left[(u_x - v_y) + i(v_x + u_y)\right].\end{aligned}$$

Thus

$$\frac{\partial f}{\partial \overline{z}} = 0$$

if and only if

$$u_x = v_y, \qquad u_y = -v_x,$$

which are precisely the Cauchy–Riemann equations. □

1.7 Power Series

Power series provide the basic local models for holomorphic functions.

Definition 1.43 (Power Series)**.** A complex power series centered at z_0 is an expression

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n, \qquad a_n \in \mathbb{C}.$$

Theorem 1.44 (Radius of Convergence)**.** *For every power series*

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

there exists a number

$$R \in [0, \infty]$$

such that the series converges absolutely whenever

$$|z - z_0| < R,$$

and diverges whenever

$$|z - z_0| > R.$$

Moreover,

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Remark 1.45. The circle

$$|z - z_0| = R$$

is called the boundary of the disc of convergence. The theorem itself does not determine what happens on this boundary; different boundary points may behave differently.

Theorem 1.46 (Differentiation of Power Series)**.** *Inside its disc of convergence,*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

is holomorphic and

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}.$$

The differentiated series has the same radius of convergence.

Consequently, a convergent power series may be differentiated repeatedly inside its disc of convergence.

Example 1.47 (The Exponential Function)**.** The complex exponential is defined by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Its radius of convergence is infinite, so e^z is entire. Term-by-term differentiation gives

$$(e^z)' = e^z.$$

The trigonometric functions may similarly be defined by

$$\cos z = \frac{e^{iz} + e^{-iz}}{2},$$

and

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Both are entire.

1.8 Analytic Functions

Definition 1.48 (Analytic Function)**.** A function f is analytic at z_0 if there exists a neighborhood of z_0 in which

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for some coefficients $a_n \in \mathbb{C}$.

Every analytic function is holomorphic, since power series may be differentiated term by term inside their discs of convergence.

The converse is much deeper.

Theorem 1.49 (Holomorphic Functions Are Analytic)**.** *If f is holomorphic in an open set Ω, then f is analytic at every point $z_0 \in \Omega$.*

Remark 1.50. The proof of this theorem will follow from Cauchy's integral formula in Chapter 3.

This is one of the fundamental differences between real and complex analysis. A real-valued function may possess infinitely many derivatives without being represented by its Taylor series. In complex analysis, holomorphicity alone forces a local power-series representation.

Thus, for complex functions,

$$\boxed{\text{analytic} \iff \text{holomorphic.}}$$

This equivalence is one of the central rigidity principles of complex analysis.

1.9 The First Rigidity Principle

The material of this chapter can be summarized by the following chain of ideas:

$$\boxed{\begin{array}{c}\text{complex derivative exists}\\ \Downarrow\\ \text{all directions of approach must agree}\\ \Downarrow\\ \text{Cauchy–Riemann equations}\\ \Downarrow\\ \text{strong restrictions on } u \text{ and } v.\end{array}}$$

Later, Cauchy's integral theorem and Cauchy's integral formula will strengthen this picture dramatically:

$$\boxed{\text{holomorphic} \Longrightarrow \text{analytic} \Longrightarrow C^\infty.}$$

The remainder of complex analysis is largely an exploration of the consequences of this rigidity.

Chapter 2

Complex Integration and Cauchy's Theorem

2 Complex Integration and Cauchy's Theorem

2.1 Parametrized Curves

Complex integration is defined along curves in the complex plane. Before introducing the integral itself, we first describe the curves over which we integrate.

Definition 2.1 (Parametrized Curve)**.** A parametrized curve is a continuous map

$$\gamma : [a, b] \longrightarrow \mathbb{C}.$$

We often write

$$\gamma(t) = x(t) + iy(t).$$

The points

$$\gamma(a), \qquad \gamma(b)$$

are called the initial point and terminal point of the curve.

Definition 2.2 (Smooth Curve)**.** A parametrized curve

$$\gamma : [a, b] \to \mathbb{C}$$

is smooth if $\gamma'(t)$ exists and is continuous on $[a, b]$, and

$$\gamma'(t) \neq 0$$

for all $t \in [a, b]$.

Definition 2.3 (Piecewise Smooth Curve)**.** A curve

$$\gamma : [a, b] \to \mathbb{C}$$

is piecewise smooth if there exists a partition

$$a = a_0 < a_1 < \cdots < a_n = b$$

such that γ is smooth on each interval

$$[a_{k-1}, a_k].$$

Definition 2.4 (Closed Curve)**.** A curve $\gamma : [a, b] \to \mathbb{C}$ is closed if

$$\gamma(a) = \gamma(b).$$

Definition 2.5 (Simple Curve)**.** A curve $\gamma : [a, b] \to \mathbb{C}$ is simple if it does not intersect itself, that is,

$$\gamma(t) \neq \gamma(s)$$

whenever $t \neq s$, except possibly for

$$\gamma(a) = \gamma(b)$$

in the closed case.

Example 2.6 (Circle)**.** The positively oriented circle centered at z_0 with radius r is parametrized by

$$\gamma(t) = z_0 + re^{it}, \qquad 0 \le t \le 2\pi.$$

The negatively oriented circle is

$$\gamma(t) = z_0 + re^{-it}, \qquad 0 \le t \le 2\pi.$$

Positive orientation means counterclockwise orientation, while negative orientation means clockwise orientation.

2.2 Equivalent Curves and Orientation

The value of a contour integral should depend on the geometric path and its orientation, not on the particular speed at which the curve is traversed.

Definition 2.7 (Equivalent Parametrizations)**.** Let

$$\gamma : [a, b] \to \mathbb{C}, \qquad \widetilde{\gamma} : [c, d] \to \mathbb{C}.$$

They are equivalent if there exists a continuously differentiable bijection

$$t = t(s) : [c, d] \to [a, b]$$

such that

$$\widetilde{\gamma}(s) = \gamma(t(s))$$

and

$$t'(s) > 0.$$

The condition $t'(s) > 0$ means that the orientation is preserved.

If instead we define

$$\gamma^-(t) = \gamma(a + b - t),$$

then γ^- traverses the same geometric curve in the opposite direction.

Remark 2.8. Reparametrization changes the speed of travel but not the oriented path. Reversing orientation, however, changes the sign of the contour integral.

2.3 Contour Integrals

Definition 2.9 (Contour Integral)**.** Let f be continuous on the image of a piecewise smooth curve

$$\gamma : [a, b] \to \mathbb{C}.$$

The contour integral of f along γ is

$$\int_\gamma f(z)\ \mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t.$$

Writing

$$\gamma(t) = x(t) + iy(t),$$

we have

$$\mathrm{d}z = \gamma'(t)\ \mathrm{d}t = \Big(x'(t) + iy'(t)\Big)\,\mathrm{d}t.$$

Thus contour integration is an ordinary real integral after a parametrization has been chosen.

Proposition 2.10 (Linearity)**.** *For constants $\alpha, \beta \in \mathbb{C}$,*

$$\int_\gamma \big(\alpha f(z) + \beta g(z)\big)\, \mathrm{d}z = \alpha \int_\gamma f(z)\, \mathrm{d}z + \beta \int_\gamma g(z)\, \mathrm{d}z.$$

Proposition 2.11 (Reversal of Orientation)**.** *If $-\gamma$ denotes the curve γ traversed in the opposite direction, then*

$$\int_{-\gamma} f(z)\, \mathrm{d}z = -\int_\gamma f(z)\, \mathrm{d}z.$$

Proof. Let

$$\widetilde{\gamma}(t) = \gamma(a + b - t).$$

Then

$$\widetilde{\gamma}'(t) = -\gamma'(a + b - t).$$

Hence

$$\int_{-\gamma} f(z)\, \mathrm{d}z = \int_a^b f(\gamma(a + b - t))\big[-\gamma'(a + b - t)\big]\, \mathrm{d}t.$$

With the change of variable

$$s = a + b - t,$$

this becomes

$$-\int_a^b f(\gamma(s))\gamma'(s)\, \mathrm{d}s.$$

□

Proposition 2.12 (Reparametrization Invariance)**.** *Equivalent parametrizations define the same contour integral.*

Proof. If

$$\widetilde{\gamma}(s) = \gamma(t(s)),$$

then

$$\widetilde{\gamma}'(s) = \gamma'(t(s))t'(s).$$

Therefore

$$\begin{aligned}\int_{\widetilde{\gamma}} f(z)\, \mathrm{d}z &= \int_c^d f(\gamma(t(s)))\gamma'(t(s))t'(s)\, \mathrm{d}s \\ &= \int_a^b f(\gamma(t))\gamma'(t)\, \mathrm{d}t \\ &= \int_\gamma f(z)\, \mathrm{d}z.\end{aligned}$$

□

2.4 Length and the Basic Estimate

Definition 2.13 (Length of a Curve)**.** For a piecewise smooth curve $\gamma : [a, b] \to \mathbb{C}$, its length is

$$L(\gamma) = \int_a^b |\gamma'(t)|\, \mathrm{d}t.$$

Theorem 2.14 (ML-Estimate)**.** *Let f be continuous on a piecewise smooth curve γ. Then*

$$\left|\int_\gamma f(z)\, \mathrm{d}z\right| \leq \sup_{z \in \gamma} |f(z)|\, L(\gamma).$$

Proof. By the triangle inequality for real integrals,

$$\begin{aligned}\left|\int_\gamma f(z)\ \mathrm{d}z\right| &= \left|\int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t\right| \\ &\le \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\ \mathrm{d}t \\ &\le \left(\sup_{z\in\gamma}|f(z)|\right)\int_a^b |\gamma'(t)|\ \mathrm{d}t.\end{aligned}$$

Hence

$$\left|\int_\gamma f(z)\ \mathrm{d}z\right| \le \sup_{z\in\gamma}|f(z)|\,L(\gamma).$$

□

Remark 2.15. This estimate is one of the most frequently used tools in complex analysis. It converts a contour integral into two simpler quantities:

$$\text{maximum size of the integrand} \times \text{length of the contour.}$$

2.5 Primitives and Path Independence

Definition 2.16 (Primitive)**.** A function F is a primitive of f in an open set Ω if

$$F'(z) = f(z)$$

for every $z \in \Omega$.

Theorem 2.17 (Fundamental Theorem for Complex Line Integrals)**.** *Suppose f has a primitive F in Ω. If γ is a piecewise smooth curve in Ω from w_1 to w_2, then*

$$\int_\gamma f(z)\ \mathrm{d}z = F(w_2) - F(w_1).$$

Proof. Let

$$\gamma : [a,b] \to \Omega.$$

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}F(\gamma(t)) = F'(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t).$$

Therefore

$$\begin{aligned}\int_\gamma f(z)\ \mathrm{d}z &= \int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t \\ &= \int_a^b \frac{\mathrm{d}}{\mathrm{d}t}F(\gamma(t))\ \mathrm{d}t \\ &= F(\gamma(b)) - F(\gamma(a)).\end{aligned}$$

□

Corollary 2.18. *If f has a primitive in Ω, then*

$$\int_\gamma f(z)\ \mathrm{d}z = 0$$

for every closed curve $\gamma \subset \Omega$.

Remark 2.19. Thus

$$\text{existence of a primitive} \implies \text{all closed contour integrals vanish.}$$

Later, under suitable assumptions on the domain, the converse will also hold.

Example 2.20. The function

$$f(z) = \frac{1}{z}$$

does not have a primitive on

$$\mathbb{C} \setminus \{0\}.$$

Indeed, let

$$\gamma(t) = e^{it}, \qquad 0 \le t \le 2\pi.$$

Then

$$\gamma'(t) = ie^{it},$$

and hence

$$\begin{aligned}\int_\gamma \frac{1}{z}\,\mathrm{d}z &= \int_0^{2\pi} \frac{1}{e^{it}} ie^{it}\,\mathrm{d}t \\ &= \int_0^{2\pi} i\,\mathrm{d}t \\ &= 2\pi i \neq 0.\end{aligned}$$

Therefore $1/z$ cannot have a primitive on $\mathbb{C} \setminus \{0\}$.

Remark 2.21. This example is fundamental. Locally,

$$\frac{1}{z}$$

does have primitives, namely branches of $\log z$. The obstruction to a global primitive is not local differentiability but the topology of the punctured plane.

2.6 Goursat's Theorem

The central fact of complex integration is that holomorphic functions have vanishing contour integrals around sufficiently simple closed curves.

Theorem 2.22 (Goursat's Theorem)**.** *Let $\Omega \subset \mathbb{C}$ be open, and let $T \subset \Omega$ be a triangle. If f is holomorphic in Ω, then*

$$\int_{\partial T} f(z)\,\mathrm{d}z = 0.$$

Remark 2.23. A striking point is that Goursat's theorem assumes only complex differentiability. No continuity of f' is required.

This is one of the first manifestations of the rigidity of holomorphic functions.

Proof. Let $T_0 = T$. Divide T_0 into four similar triangles by joining the midpoints of its sides. Since

$$\int_{\partial T_0} f(z)\,\mathrm{d}z$$

is the sum of the integrals over the boundaries of the four smaller triangles, at least one of them, say T_1, satisfies

$$\left|\int_{\partial T_1} f(z)\,\mathrm{d}z\right| \ge \frac{1}{4}\left|\int_{\partial T_0} f(z)\,\mathrm{d}z\right|.$$

Repeat the construction. We obtain nested triangles

$$T_0 \supset T_1 \supset T_2 \supset \cdots$$

such that

$$\left|\int_{\partial T_n} f(z)\ \mathrm{d}z\right| \geq \frac{1}{4^n}\left|\int_{\partial T_0} f(z)\ \mathrm{d}z\right|.$$

The diameters satisfy

$$\operatorname{diam}(T_n) = 2^{-n}\operatorname{diam}(T_0),$$

so by the nested compact set theorem there exists a unique point

$$z_0 \in \bigcap_{n=0}^{\infty} T_n.$$

Since f is differentiable at z_0,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \eta(z)(z - z_0),$$

where

$$\eta(z) \to 0 \qquad \text{as } z \to z_0.$$

The first two terms have primitives, hence their integrals around ∂T_n vanish. Thus

$$\int_{\partial T_n} f(z)\ \mathrm{d}z = \int_{\partial T_n} \eta(z)(z - z_0)\ \mathrm{d}z.$$

Let

$$M_n = \sup_{z \in T_n} |\eta(z)|.$$

Then $M_n \to 0$. By the ML-estimate,

$$\left|\int_{\partial T_n} f(z)\ \mathrm{d}z\right| \leq M_n \operatorname{diam}(T_n) L(\partial T_n).$$

Both $\operatorname{diam}(T_n)$ and $L(\partial T_n)$ are of order 2^{-n}, so

$$\left|\int_{\partial T_n} f(z)\ \mathrm{d}z\right| \leq C M_n 4^{-n}$$

for some constant $C > 0$.

Combining this with

$$\frac{1}{4^n}\left|\int_{\partial T_0} f(z)\ \mathrm{d}z\right| \leq \left|\int_{\partial T_n} f(z)\ \mathrm{d}z\right|,$$

we obtain

$$\left|\int_{\partial T_0} f(z)\ \mathrm{d}z\right| \leq C M_n.$$

Letting $n \to \infty$ gives

$$\int_{\partial T_0} f(z)\ \mathrm{d}z = 0.$$

□

Corollary 2.24. *Let $R \subset \Omega$ be a rectangle. If f is holomorphic in Ω, then*

$$\int_{\partial R} f(z)\ \mathrm{d}z = 0.$$

Remark 2.25. A rectangle can be divided into two triangles. Applying Goursat's theorem to each triangle and cancelling the integral along the common diagonal gives the result.

2.7 Local Existence of Primitives

Theorem 2.26 (Local Existence of a Primitive)**.** *A holomorphic function in an open disc has a primitive.*

Proof. Let

$$D = D_r(z_0),$$

and let f be holomorphic in D. Fix $a \in D$. For $z \in D$, define

$$F(z) = \int_{\gamma_{a,z}} f(w)\ \mathrm{d}w,$$

where $\gamma_{a,z}$ is the polygonal path consisting of straight line segments from a to z lying inside the disc.

Because a disc is convex, the relevant triangles remain inside D. Goursat's theorem implies that the integral is independent of the chosen polygonal path.

Now let h be small enough that the segment from z to $z+h$ lies inside D. Then

$$F(z+h) - F(z) = \int_z^{z+h} f(w)\ \mathrm{d}w.$$

Hence

$$\frac{F(z+h)-F(z)}{h} - f(z) = \frac{1}{h}\int_z^{z+h} \Big(f(w) - f(z)\Big)\ \mathrm{d}w.$$

Parametrize the segment by

$$w = z + th, \qquad 0 \le t \le 1.$$

Then

$$\frac{F(z+h)-F(z)}{h} - f(z) = \int_0^1 \Big(f(z+th) - f(z)\Big)\ \mathrm{d}t.$$

Since f is continuous,

$$f(z+th) \to f(z)$$

uniformly for $0 \le t \le 1$ as $h \to 0$. Therefore

$$F'(z) = f(z).$$

□

Corollary 2.27 (Cauchy's Theorem in a Disc)**.** *If f is holomorphic in a disc D, then*

$$\int_\gamma f(z)\ \mathrm{d}z = 0$$

for every closed piecewise smooth curve $\gamma \subset D$.

Proof. By the previous theorem, f has a primitive in D. Therefore every closed contour integral vanishes. □

2.8 Homotopy of Curves

The vanishing of contour integrals is closely connected with the topology of the domain.

Definition 2.28 (Homotopic Curves)**.** Let

$$\gamma_0, \gamma_1 : [a, b] \to \Omega$$

be two curves with the same initial and terminal points:

$$\gamma_0(a) = \gamma_1(a) = \alpha,$$

and

$$\gamma_0(b) = \gamma_1(b) = \beta.$$

The curves are homotopic in Ω if there exists a continuous map

$$H : [0, 1] \times [a, b] \to \Omega$$

such that

$$H(0, t) = \gamma_0(t), \qquad H(1, t) = \gamma_1(t),$$

and

$$H(s, a) = \alpha, \qquad H(s, b) = \beta$$

for every $s \in [0, 1]$.

Intuitively, γ_0 can be continuously deformed into γ_1 without leaving the domain and without moving the endpoints.

Theorem 2.29 (Homotopy Invariance)**.** *Let f be holomorphic in an open set Ω. If γ_0 and γ_1 are homotopic in Ω with fixed endpoints, then*

$$\int_{\gamma_0} f(z)\, \mathrm{d}z = \int_{\gamma_1} f(z)\, \mathrm{d}z.$$

Remark 2.30. This theorem says that for holomorphic functions, the contour integral is unchanged under continuous deformation of the path, as long as the deformation does not cross a singularity or leave the domain.

2.9 Simply Connected Domains

Definition 2.31 (Simply Connected Domain)**.** A domain $\Omega \subset \mathbb{C}$ is simply connected if any two curves in Ω having the same endpoints are homotopic with fixed endpoints.

Equivalently, every closed curve in Ω can be continuously contracted to a point inside Ω.

Example 2.32. The following domains are simply connected:

$$\mathbb{C}, \qquad D_r(z_0), \qquad \mathbb{H} = \{z : \Im z > 0\}.$$

Example 2.33. The punctured plane

$$\mathbb{C} \setminus \{0\}$$

is not simply connected.

A circle surrounding 0 cannot be continuously contracted to a point without crossing the missing point 0.

Theorem 2.34 (Primitive on a Simply Connected Domain)**.** *Every holomorphic function on a simply connected domain has a primitive.*

Proof. Fix $z_0 \in \Omega$. For $z \in \Omega$, define

$$F(z) = \int_{\gamma_z} f(w)\ \mathrm{d}w,$$

where γ_z is any path in Ω from z_0 to z.

If γ_1 and γ_2 are two such paths, then they are homotopic with fixed endpoints because Ω is simply connected. By homotopy invariance,

$$\int_{\gamma_1} f(w)\ \mathrm{d}w = \int_{\gamma_2} f(w)\ \mathrm{d}w.$$

Therefore F is well defined.

The same local argument used for a disc shows that

$$F'(z) = f(z).$$

Hence F is a primitive of f. □

Corollary 2.35 (Cauchy's Theorem in Simply Connected Domains)**.** *Let f be holomorphic in a simply connected domain Ω. Then for every closed curve $\gamma \subset \Omega$,*

$$\int_\gamma f(z)\ \mathrm{d}z = 0.$$

2.10 Equivalent Forms of Cauchy's Theorem

For a holomorphic function f on a domain Ω, the following statements are closely related:

$$\boxed{\begin{array}{c} f \text{ has a primitive} \\ \Downarrow \\ \displaystyle\int_\gamma f(z)\ \mathrm{d}z = 0 \quad \text{for every closed curve } \gamma \\ \Downarrow \\ \text{integrals are path independent.} \end{array}}$$

On a simply connected domain, these properties hold automatically for every holomorphic function.

Remark 2.36. The example

$$f(z) = \frac{1}{z}$$

on

$$\mathbb{C} \setminus \{0\}$$

shows why the topology of the domain matters. The function is holomorphic everywhere on the punctured plane, yet

$$\int_{|z|=1} \frac{1}{z}\ \mathrm{d}z = 2\pi i.$$

Thus holomorphicity alone does not imply vanishing of every closed contour integral unless the domain has the appropriate topology.

2.11 Main Idea of the Chapter

The main logical structure of this chapter is

holomorphicity
$\Downarrow$
Goursat's theorem
$\Downarrow$
local primitives
$\Downarrow$
vanishing of closed contour integrals.

When the domain is simply connected, the local result becomes global:

holomorphic on a simply connected domain $\Longrightarrow$ primitive exists $\Longrightarrow \oint_\gamma f(z)\,\mathrm{d}z = 0.$

This is the foundation for Cauchy's integral formula, which will convert contour integration into precise information about the values and derivatives of holomorphic functions inside a domain.

Chapter 3

Cauchy's Integral Formula and Its Consequences

3 Cauchy's Integral Formula and Its Consequences

3.1 Cauchy's Integral Formula

Cauchy's theorem states that the integral of a holomorphic function around a closed contour vanishes under suitable conditions. Cauchy's integral formula goes much further: it recovers the value of a holomorphic function inside a circle entirely from its values on the boundary.

Theorem 3.1 (Cauchy's Integral Formula)**.** *Let f be holomorphic in an open set containing the closed disc*

$$\overline{D}_r(z_0) = \{z \in \mathbb{C} : |z - z_0| \le r\}.$$

Let

$$C = \partial D_r(z_0)$$

be positively oriented. Then for every $z \in D_r(z_0)$,

$$\boxed{f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, \mathrm{d}\zeta.}$$

Proof. Fix $z \in D_r(z_0)$. Choose a small circle

$$C_\varepsilon = \{\zeta : |\zeta - z| = \varepsilon\}$$

contained in $D_r(z_0)$.

The function

$$\frac{f(\zeta)}{\zeta - z}$$

is holomorphic in the region between C and C_ε. By Cauchy's theorem,

$$\int_C \frac{f(\zeta)}{\zeta - z}\, \mathrm{d}\zeta = \int_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z}\, \mathrm{d}\zeta.$$

Write

$$f(\zeta) = f(z) + \Big(f(\zeta) - f(z)\Big).$$

Then

$$\begin{aligned}\int_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z}\, \mathrm{d}\zeta &= f(z) \int_{C_\varepsilon} \frac{1}{\zeta - z}\, \mathrm{d}\zeta \\ &\quad + \int_{C_\varepsilon} \frac{f(\zeta) - f(z)}{\zeta - z}\, \mathrm{d}\zeta.\end{aligned}$$

Parametrize

$$\zeta = z + \varepsilon e^{it}, \qquad 0 \le t \le 2\pi.$$

Then

$$\mathrm{d}\zeta = i\varepsilon e^{it}\, \mathrm{d}t,$$

so

$$\begin{aligned}\int_{C_\varepsilon} \frac{1}{\zeta - z}\, \mathrm{d}\zeta &= \int_0^{2\pi} \frac{i\varepsilon e^{it}}{\varepsilon e^{it}}\, \mathrm{d}t \\ &= 2\pi i.\end{aligned}$$

For the second term, the ML-estimate gives

$$\left| \int_{C_\varepsilon} \frac{f(\zeta) - f(z)}{\zeta - z} \, \mathrm{d}\zeta \right| \leq 2\pi \sup_{|\zeta - z| = \varepsilon} |f(\zeta) - f(z)|.$$

Since f is continuous, the right-hand side tends to 0 as $\varepsilon \to 0$.

Therefore

$$\int_C \frac{f(\zeta)}{\zeta - z} \, \mathrm{d}\zeta = 2\pi i f(z),$$

which proves the formula. □

Remark 3.2. Cauchy's integral formula is a striking local-to-global statement:

$$\boxed{\text{boundary values of } f \Longrightarrow \text{interior values of } f.}$$

A function defined on a two-dimensional region is completely controlled by its values on a one-dimensional surrounding curve.

Example 3.3. Let

$$C = \{z : |z| = 2\}$$

with positive orientation. Then

$$\int_C \frac{e^z}{z - 1} \, \mathrm{d}z = 2\pi i e.$$

Indeed, e^z is entire and 1 lies inside C, so Cauchy's integral formula gives the result immediately.

Example 3.4. Let

$$C = \{z : |z| = 1\}.$$

Then

$$\int_C \frac{e^z}{z - 2} \, \mathrm{d}z = 0.$$

Although the integrand has a singularity at $z = 2$, that singularity lies outside the contour. Hence the integrand is holomorphic on and inside C, so Cauchy's theorem applies.

3.2 Higher Derivatives

One of the strongest consequences of Cauchy's integral formula is that a holomorphic function automatically possesses derivatives of every order.

Theorem 3.5 (Cauchy's Integral Formula for Derivatives)**.** *Under the assumptions of Cauchy's integral formula,*

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} \, \mathrm{d}\zeta, \qquad n \geq 0.$$

In particular,

$$f'(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^2} \, \mathrm{d}\zeta.$$

Remark 3.6. This result immediately gives

$$\boxed{f \text{ holomorphic} \Longrightarrow f \in C^\infty.}$$

This phenomenon has no analogue in ordinary real differentiability: complex differentiability once already forces differentiability infinitely many times.

Example 3.7. Let C be a positively oriented circle containing 0 in its interior. Then

$$\int_C \frac{e^z}{z^4}\,\mathrm{d}z$$

can be evaluated without parametrizing C.

Since

$$\frac{e^z}{z^4} = \frac{f(z)}{z^{3+1}}$$

with $f(z) = e^z$, Cauchy's formula for derivatives gives

$$\int_C \frac{e^z}{z^4}\,\mathrm{d}z = \frac{2\pi i}{3!} f^{(3)}(0) = \frac{2\pi i}{6}.$$

Therefore

$$\int_C \frac{e^z}{z^4}\,\mathrm{d}z = \frac{\pi i}{3}.$$

3.3 Cauchy's Inequalities

The derivative formula not only proves the existence of derivatives; it also bounds their size.

Theorem 3.8 (Cauchy's Inequalities)**.** *Suppose f is holomorphic in an open set containing*

$$\overline{D}_R(z_0).$$

Let

$$M_R = \sup_{|z-z_0|=R} |f(z)|.$$

Then for every $n \geq 0$,

$$\boxed{|f^{(n)}(z_0)| \leq \frac{n! M_R}{R^n}.}$$

Proof. By Cauchy's integral formula for derivatives,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{|\zeta - z_0| = R} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,\mathrm{d}\zeta.$$

Hence

$$\begin{aligned} |f^{(n)}(z_0)| &\leq \frac{n!}{2\pi} \frac{M_R}{R^{n+1}} (2\pi R) \\ &= \frac{n! M_R}{R^n}. \end{aligned}$$

□

Remark 3.9. The significance is not merely that derivatives exist. Their sizes are controlled quantitatively by the size of the original function.

Thus holomorphic functions cannot oscillate arbitrarily rapidly while remaining uniformly small.

3.4 Holomorphic Functions Are Analytic

We can now prove the result stated without proof in Chapter 1.

Theorem 3.10 (Power-Series Expansion)**.** *Let f be holomorphic in a disc*

$$D_R(z_0).$$

Then

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \qquad |z - z_0| < R,$$

where

$$a_n = \frac{f^{(n)}(z_0)}{n!}.$$

Equivalently,

$$\boxed{f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.}$$

Proof. Fix $r < R$, and let

$$|z - z_0| < r.$$

By Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} \, \mathrm{d}\zeta.$$

Write

$$\zeta - z = (\zeta - z_0) - (z - z_0).$$

Therefore

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0} \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}}.$$

Since

$$\left| \frac{z - z_0}{\zeta - z_0} \right| < 1,$$

we may use the geometric series

$$\frac{1}{1 - w} = \sum_{n=0}^{\infty} w^n.$$

Thus

$$\frac{1}{\zeta - z} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}.$$

Substituting this into Cauchy's formula gives

$$f(z) = \sum_{n=0}^{\infty} \left[\frac{1}{2\pi i} \int_{|\zeta - z_0| = r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, \mathrm{d}\zeta \right] (z - z_0)^n.$$

By Cauchy's formula for derivatives,

$$\frac{1}{2\pi i} \int_{|\zeta - z_0| = r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, \mathrm{d}\zeta = \frac{f^{(n)}(z_0)}{n!}.$$

Hence

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

□

Corollary 3.11. *A function is holomorphic if and only if it is analytic.*

Thus

$$\boxed{\text{holomorphic} \iff \text{analytic.}}$$

Remark 3.12. This equivalence is one of the fundamental rigidity principles of complex analysis.

In real analysis,

$$C^\infty \not\Longrightarrow \text{analytic.}$$

In complex analysis, however,

$$\text{one complex derivative locally} \Longrightarrow \text{a convergent power series.}$$

3.5 Liouville's Theorem

Cauchy's inequalities have a remarkably strong global consequence.

Theorem 3.13 (Liouville's Theorem)**.** *If f is entire and bounded, then f is constant.*

Proof. Suppose

$$|f(z)| \leq M$$

for every $z \in \mathbb{C}$.

Fix $z_0 \in \mathbb{C}$. Since f is entire, Cauchy's inequality can be applied on every circle

$$|z - z_0| = R.$$

Thus

$$|f'(z_0)| \leq \frac{M}{R}.$$

Letting

$$R \to \infty$$

gives

$$f'(z_0) = 0.$$

Since z_0 was arbitrary,

$$f' \equiv 0.$$

Therefore f is constant. □

Remark 3.14. The proof illustrates the power of complex analysis very clearly.

The assumption that f is bounded is global, while Cauchy's inequality is local. Because an entire function is defined on discs of arbitrarily large radius, the local estimate can be applied with

$$R \to \infty,$$

forcing every derivative to vanish.

3.6 The Fundamental Theorem of Algebra

Liouville's theorem gives a short proof of one of the fundamental results of algebra.

Theorem 3.15 (Fundamental Theorem of Algebra)**.** *Every nonconstant polynomial*

$$P(z) = a_n z^n + \cdots + a_1 z + a_0, \qquad a_n \neq 0,$$

has at least one zero in $\mathbb{C}$*.*

Proof. Suppose, to the contrary, that

$$P(z) \neq 0$$

for every $z \in \mathbb{C}$. Then

$$f(z) = \frac{1}{P(z)}$$

is entire.

As $|z| \to \infty$,

$$|P(z)| \to \infty,$$

and therefore

$$\frac{1}{|P(z)|} \to 0.$$

Hence $1/P$ is bounded outside a sufficiently large disc.

Inside that closed disc, $1/P$ is continuous and therefore bounded by compactness. Thus $1/P$ is bounded on all of $\mathbb{C}$.

Liouville's theorem implies that $1/P$ is constant, so P is constant, a contradiction. □

Corollary 3.16. *Every polynomial of degree* $n \geq 1$ *can be written as*

$$P(z) = a_n(z - w_1) \cdots (z - w_n),$$

where

$$w_1, \ldots, w_n \in \mathbb{C},$$

with roots repeated according to their multiplicities.

3.7 Zeros of Holomorphic Functions

Zeros of holomorphic functions are much more rigid than zeros of general continuous or smooth functions.

Theorem 3.17 (Local Form Near a Zero)**.** *Let* f *be holomorphic in a neighborhood of* z_0*, and suppose*

$$f(z_0) = 0.$$

If f *is not identically zero near* z_0*, then there exists a unique integer* $m \geq 1$ *and a holomorphic function* g *such that*

$$f(z) = (z - z_0)^m g(z),$$

where

$$g(z_0) \neq 0.$$

Proof. Since f is analytic,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

Since $f(z_0) = 0$,

$$a_0 = 0.$$

Because f is not identically zero, there exists a smallest integer $m \geq 1$ for which

$$a_m \neq 0.$$

Therefore

$$f(z) = (z - z_0)^m \left[a_m + a_{m+1}(z - z_0) + \cdots \right].$$

Define

$$g(z) = a_m + a_{m+1}(z - z_0) + \cdots .$$

Then g is holomorphic and

$$g(z_0) = a_m \neq 0.$$

□

Definition 3.18 (Order of a Zero)**.** In the representation

$$f(z) = (z - z_0)^m g(z), \qquad g(z_0) \neq 0,$$

the integer m is called the order or multiplicity of the zero z_0.

If

$$m = 1,$$

then z_0 is called a simple zero.

Corollary 3.19 (Zeros Are Isolated)**.** *If f is holomorphic and not identically zero, then every zero of f is isolated.*

Proof. If

$$f(z) = (z - z_0)^m g(z)$$

with $g(z_0) \neq 0$, then by continuity there exists $r > 0$ such that

$$g(z) \neq 0$$

for

$$|z - z_0| < r.$$

Therefore z_0 is the only zero of f in a sufficiently small neighborhood of z_0. □

3.8 The Identity Theorem

Theorem 3.20 (Identity Theorem)**.** *Let Ω be a connected open set, and let f be holomorphic in Ω. Suppose f vanishes on a sequence of distinct points*

$$z_1, z_2, \ldots$$

having a limit point in Ω. Then

$$f \equiv 0$$

in Ω.

Proof. Let $z_0 \in \Omega$ be a limit point of the zeros.

If f were not identically zero, then by the local form near a zero, z_0 would be an isolated zero. But every neighborhood of z_0 contains some $z_n \neq z_0$ with

$$f(z_n) = 0.$$

This contradicts isolation of zeros.

Hence $f \equiv 0$ in a neighborhood of z_0. Since Ω is connected, analytic continuation through overlapping discs gives

$$f \equiv 0$$

throughout Ω. □

Corollary 3.21 (Uniqueness Theorem)**.** *Let f and g be holomorphic in a connected open set Ω. Suppose*

$$f(z_n) = g(z_n)$$

for a sequence of distinct points z_n having a limit point in Ω. Then

$$f(z) = g(z)$$

for every $z \in \Omega$.

Proof. Apply the identity theorem to

$$h = f - g.$$

□

Remark 3.22. The assumption that the limit point lies *inside* the domain is essential.

For example,

$$f(z) = \sin \frac{\pi}{z}$$

is holomorphic on

$$\mathbb{C} \setminus \{0\}$$

and has zeros

$$z_n = \frac{1}{n}.$$

These zeros satisfy

$$z_n \to 0,$$

but 0 is not in the domain. Thus f is certainly not identically zero.

Therefore accumulation of zeros at a boundary point does not imply identical vanishing.

Remark 3.23. The identity theorem expresses another major rigidity property:

a holomorphic function is determined by its values on any set with an interior accumulation point.

This is much stronger than what is true for ordinary smooth functions.

3.9 The Maximum Modulus Principle

The modulus of a nonconstant holomorphic function cannot have a genuine maximum at an interior point.

Theorem 3.24 (Maximum Modulus Principle)**.** *Let f be holomorphic and nonconstant in a region Ω. Then*

$$|f|$$

cannot attain a maximum at an interior point of Ω.

Proof. Suppose that $z_0 \in \Omega$ is an interior point at which

$$|f(z)| \leq |f(z_0)|$$

in a neighborhood of z_0.

Choose $r > 0$ such that

$$\overline{D}_r(z_0) \subset \Omega.$$

By Cauchy's integral formula,

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) \, \mathrm{d}t.$$

Therefore

$$\begin{aligned} |f(z_0)| &\leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})| \, \mathrm{d}t \\ &\leq |f(z_0)|. \end{aligned}$$

Equality throughout forces

$$|f(z_0 + re^{it})| = |f(z_0)|$$

for every t. Repeating the argument locally implies that $|f|$ is constant in a neighborhood of z_0, and consequently f is constant there. By the identity theorem, f is constant on Ω, a contradiction. □

Corollary 3.25. *Let Ω be a bounded region, and suppose f is holomorphic in Ω and continuous on*

$$\overline{\Omega}.$$

Then

$$\max_{z \in \overline{\Omega}} |f(z)| = \max_{z \in \partial\Omega} |f(z)|.$$

Remark 3.26. This is another form of the principle

boundary controls interior.

For a nonconstant holomorphic function, the largest value of $|f|$ on a compact region must occur on the boundary.

3.10 The Open Mapping Theorem

Theorem 3.27 (Open Mapping Theorem)**.** *Let f be holomorphic and nonconstant in a region Ω. Then f is an open map: if $U \subset \Omega$ is open, then*

$$f(U)$$

is open in $\mathbb{C}$.

Remark 3.28. A nonconstant holomorphic function cannot collapse an open two-dimensional region into a curve or a set with empty interior.

Thus holomorphic functions are rigid analytically, but locally they also behave like genuinely two-dimensional mappings.

Corollary 3.29. *If f is holomorphic and nonconstant in Ω, then*

$$|f|$$

cannot attain a local minimum at $z_0 \in \Omega$ whenever

$$f(z_0) \neq 0.$$

Proof. If $f(z_0) \neq 0$, then $1/f$ is holomorphic near z_0. A local minimum of $|f|$ would give a local maximum of

$$\left|\frac{1}{f}\right|,$$

contradicting the maximum modulus principle. □

3.11 Morera's Theorem

Cauchy's theorem states

$$\text{holomorphic} \implies \text{vanishing contour integrals.}$$

Morera's theorem gives an important converse.

Theorem 3.30 (Morera's Theorem)**.** *Let f be continuous in an open set Ω. Suppose*

$$\int_{\partial T} f(z)\ \mathrm{d}z = 0$$

for every triangle T whose interior and boundary lie in Ω. Then f is holomorphic in Ω.

Proof. Fix $z_0 \in \Omega$, and choose a disc

$$D \subset \Omega$$

centered at z_0.

For $z \in D$, define

$$F(z) = \int_{z_0}^{z} f(w)\ \mathrm{d}w,$$

where the integral is taken along any polygonal path in D.

The assumption that all triangular integrals vanish implies that this integral is path independent. As in the proof of local existence of primitives,

$$F'(z) = f(z).$$

Thus F is holomorphic.

Since holomorphic functions are infinitely differentiable,

$$f = F'$$

is holomorphic. Since z_0 was arbitrary, f is holomorphic throughout Ω. □

Remark 3.31. The continuity assumption is important in the standard form of Morera's theorem.

Together with Cauchy's theorem, Morera's theorem gives the conceptual correspondence

$$\boxed{\text{holomorphicity} \quad \longleftrightarrow \quad \text{local vanishing of contour integrals,}}$$

provided the appropriate hypotheses are imposed.

3.12 Uniform Limits of Holomorphic Functions

A major advantage of holomorphic functions is that holomorphicity is stable under locally uniform convergence.

Theorem 3.32 (Weierstrass Convergence Theorem)**.** *Let $\{f_n\}$ be holomorphic in an open set Ω. Suppose*

$$f_n \longrightarrow f$$

uniformly on every compact subset of Ω. Then f is holomorphic in Ω.

Proof. Since the convergence is uniform on compact subsets, f is continuous.

Let T be any triangle whose interior and boundary lie in Ω. Uniform convergence on ∂T gives

$$\int_{\partial T} f(z)\, \mathrm{d}z = \lim_{n\to\infty} \int_{\partial T} f_n(z)\, \mathrm{d}z.$$

Since every f_n is holomorphic,

$$\int_{\partial T} f_n(z)\, \mathrm{d}z = 0.$$

Hence

$$\int_{\partial T} f(z)\, \mathrm{d}z = 0.$$

Morera's theorem implies that f is holomorphic. □

Theorem 3.33 (Convergence of Derivatives)**.** *Under the same assumptions,*

$$f_n' \longrightarrow f'$$

uniformly on every compact subset of Ω.

More generally, for every $k \geq 1$,

$$f_n^{(k)} \longrightarrow f^{(k)}$$

uniformly on compact subsets of Ω.

Proof. Let $K \subset \Omega$ be compact. Choose a positively oriented circle $C \subset \Omega$ surrounding a neighborhood of each point under consideration.

By Cauchy's formula,

$$f_n'(z) - f'(z) = \frac{1}{2\pi i} \int_C \frac{f_n(\zeta) - f(\zeta)}{(\zeta - z)^2}\, \mathrm{d}\zeta.$$

The distance between the relevant compact set and C is positive. Therefore the ML-estimate gives

$$|f_n'(z) - f'(z)| \leq C_K \sup_{\zeta \in C} |f_n(\zeta) - f(\zeta)|$$

for a constant C_K independent of n.

The right-hand side tends uniformly to zero. Hence

$$f_n' \to f'$$

uniformly on K. Higher derivatives follow in the same way. □

Remark 3.34. This is much stronger than the corresponding statement for real differentiable functions.

For real functions, uniform convergence

$$f_n \to f$$

does not generally imply convergence of derivatives or even differentiability of the limit.

For holomorphic functions, locally uniform convergence automatically preserves all complex derivatives.

3.13 Holomorphic Dependence Under an Integral

Theorem 3.35. *Let $\Omega \subset \mathbb{C}$ be open, and suppose*

$$F : \Omega \times [0,1] \to \mathbb{C}$$

satisfies:

1. *for every $s \in [0,1]$, the function*

$$z \longmapsto F(z,s)$$

   is holomorphic in Ω;

2. *F is continuous on*

$$\Omega \times [0,1].$$

Then

$$f(z) = \int_0^1 F(z,s)\ \mathrm{d}s$$

is holomorphic in Ω.

Proof. Let $T \subset \Omega$ be a triangle. By continuity, the order of integration may be interchanged:

$$\begin{aligned} \int_{\partial T} f(z)\ \mathrm{d}z &= \int_{\partial T} \int_0^1 F(z,s)\ \mathrm{d}s\ \mathrm{d}z \\ &= \int_0^1 \left(\int_{\partial T} F(z,s)\ \mathrm{d}z \right) \mathrm{d}s. \end{aligned}$$

For every fixed s, the function $F(\cdot, s)$ is holomorphic, so Cauchy's theorem gives

$$\int_{\partial T} F(z,s)\ \mathrm{d}z = 0.$$

Therefore

$$\int_{\partial T} f(z)\ \mathrm{d}z = 0.$$

Morera's theorem implies that f is holomorphic. □

3.14 Symmetry and Schwarz Reflection

Suppose a domain is symmetric with respect to the real axis. Complex conjugation then produces a natural way to reflect holomorphic functions across the real axis.

Let

$$\Omega^+ = \{z \in \Omega : \Im z > 0\},$$
$$\Omega^- = \{z \in \Omega : \Im z < 0\},$$

and

$$I = \Omega \cap \mathbb{R}.$$

Theorem 3.36 (Symmetry Principle)**.** *Suppose f^+ is holomorphic in Ω^+ and f^- is holomorphic in Ω^-, and both extend continuously to I. If*

$$f^+(x) = f^-(x)$$

for every $x \in I$, then the function

$$F(z) = \begin{cases} f^+(z), & z \in \Omega^+, \\ f^+(z) = f^-(z), & z \in I, \\ f^-(z), & z \in \Omega^- \end{cases}$$

is holomorphic in Ω.

Theorem 3.37 (Schwarz Reflection Principle)**.** *Suppose f is holomorphic in Ω^+, extends continuously to I, and*

$$f(x) \in \mathbb{R}$$

for every $x \in I$.

Then f extends holomorphically to the reflected domain by defining

$$F(z) = \begin{cases} f(z), & \Im z \geq 0, \\ \overline{f(\overline{z})}, & \Im z < 0. \end{cases}$$

Remark 3.38. The formula

$$F(z) = \overline{f(\overline{z})}$$

may initially appear anti-holomorphic because complex conjugation appears twice. The two conjugations cancel each other's orientation reversal, and the resulting reflected function is holomorphic.

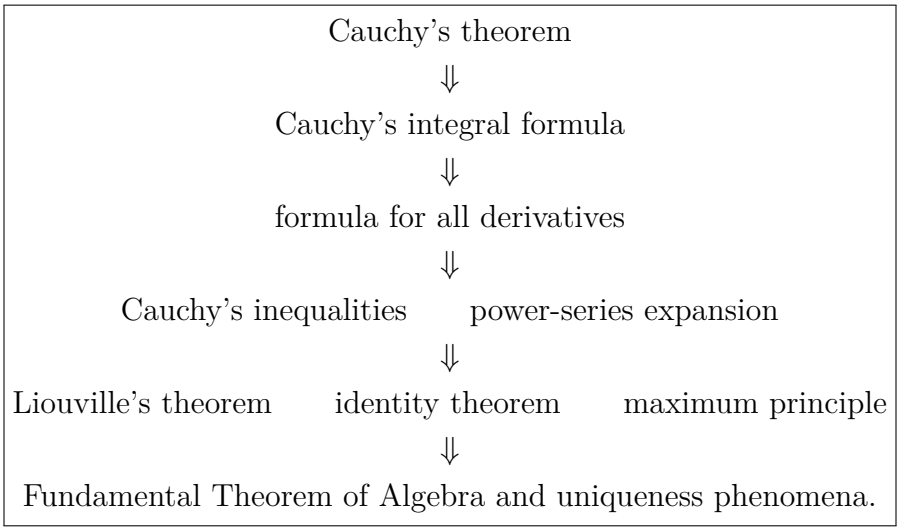
The condition

$$f(x) \in \mathbb{R}$$

on the real axis ensures that the upper and lower definitions agree on the boundary.

3.15 The Rigidity of Holomorphic Functions

The results of this chapter form one of the central implication chains in complex analysis:



The most important conceptual conclusion is

$$\boxed{\text{holomorphic} \Longleftrightarrow \text{analytic.}}$$

Thus complex differentiability is not merely a local first-order regularity condition. It controls all derivatives, all local Taylor coefficients, the possible distribution of zeros, and even much of the global behavior of the function.

Chapter 4

Zeros, Singularities and Laurent Series

4 Zeros, Singularities and Laurent Series

4.1 Zeros of Holomorphic Functions

We have already seen that zeros of a nonzero holomorphic function are isolated. We now describe more precisely the local structure of a holomorphic function near a zero.

Definition 4.1 (Zero)**.** Let f be holomorphic in a neighborhood of z_0. The point z_0 is called a zero of f if

$$f(z_0) = 0.$$

Theorem 4.2 (Local Form Near a Zero)**.** *Let f be holomorphic in a neighborhood of z_0, and suppose that $f(z_0) = 0$ but f does not vanish identically. Then there exists a neighborhood U of z_0, a nonvanishing holomorphic function g on U, and a unique integer $n \geq 1$ such that*

$$f(z) = g(z)(z - z_0)^n, \qquad z \in U.$$

Definition 4.3 (Order of a Zero)**.** In the factorization

$$f(z) = g(z)(z - z_0)^n, \qquad g(z_0) \neq 0,$$

the integer n is called the order or multiplicity of the zero z_0.

If

$$n = 1,$$

then z_0 is called a simple zero.

Remark 4.4. The local factorization separates the two pieces of information near a zero:

$$f(z) = \underbrace{(z - z_0)^n}_{\text{vanishing behavior}} \underbrace{g(z)}_{\text{nonzero holomorphic part}}.$$

Thus, up to multiplication by a nonvanishing holomorphic function, every zero behaves exactly like a power of $z - z_0$.

Corollary 4.5. *If f is holomorphic and not identically zero in a region, then its zeros are isolated.*

4.2 Isolated Singularities

Holomorphic functions may fail to be defined or holomorphic at isolated points. Such exceptional points are called isolated singularities.

Definition 4.6 (Deleted Neighborhood)**.** A deleted neighborhood of z_0 is a set of the form

$$0 < |z - z_0| < r, \qquad r > 0.$$

Definition 4.7 (Isolated Singularity)**.** Suppose f is holomorphic in a deleted neighborhood

$$0 < |z - z_0| < r.$$

Then z_0 is called an isolated singularity of f.

There are three possible types of isolated singularities:

removable singularity,
pole,
essential singularity.

4.3 Removable Singularities

Definition 4.8 (Removable Singularity)**.** Let f be holomorphic in a deleted neighborhood of z_0. The point z_0 is called a removable singularity if one can define $f(z_0)$ so that the resulting function is holomorphic in a full neighborhood of z_0.

Example 4.9. The function

$$f(z) = \frac{\sin z}{z}$$

is holomorphic on

$$\mathbb{C} \setminus \{0\}.$$

Since

$$\lim_{z \to 0} \frac{\sin z}{z} = 1,$$

we can define

$$f(0) = 1.$$

The resulting function is entire. Hence 0 is a removable singularity.

Theorem 4.10 (Riemann's Theorem on Removable Singularities)**.** *Let f be holomorphic in*

$$0 < |z - z_0| < r.$$

If f is bounded near z_0, then z_0 is a removable singularity.

Proof. Assume

$$|f(z)| \leq M$$

for

$$0 < |z - z_0| < r.$$

Define

$$g(z) = \begin{cases} (z - z_0)^2 f(z), & z \neq z_0, \\ 0, & z = z_0. \end{cases}$$

For $z \neq z_0$, the function g is holomorphic. Moreover,

$$\left| \frac{g(z) - g(z_0)}{z - z_0} \right| = |(z - z_0) f(z)| \leq M|z - z_0| \longrightarrow 0.$$

Therefore

$$g'(z_0) = 0,$$

so g is holomorphic at z_0.

Since

$$g(z_0) = 0 \qquad \text{and} \qquad g'(z_0) = 0,$$

the Taylor expansion of g has the form

$$g(z) = (z - z_0)^2 h(z)$$

for some holomorphic function h. Thus for $z \neq z_0$,

$$f(z) = h(z).$$

Defining

$$f(z_0) = h(z_0)$$

gives a holomorphic extension across z_0. □

Remark 4.11. Thus, for an isolated singularity,

$$\boxed{f \text{ bounded near } z_0 \Longrightarrow z_0 \text{ removable.}}$$

The singularity may look problematic only because the original formula was not defined at z_0; analytically, nothing singular is actually happening.

4.4 Poles

Definition 4.12 (Pole)**.** Let f be holomorphic in a deleted neighborhood of z_0. We say that f has a pole at z_0 if

$$\frac{1}{f(z)}$$

extends holomorphically to z_0 and has a zero there.

Equivalently,

$$|f(z)| \longrightarrow \infty \qquad \text{as } z \to z_0.$$

Theorem 4.13 (Local Form Near a Pole)**.** *The function f has a pole at z_0 if and only if, in a neighborhood of z_0,*

$$f(z) = \frac{h(z)}{(z - z_0)^n},$$

where $n \geq 1$ and h is holomorphic with

$$h(z_0) \neq 0.$$

Definition 4.14 (Order of a Pole)**.** In the representation

$$f(z) = \frac{h(z)}{(z - z_0)^n}, \qquad h(z_0) \neq 0,$$

the integer n is called the order of the pole.

If

$$n = 1,$$

the pole is called simple.

Example 4.15. The function

$$f(z) = \frac{1}{(z-1)^3}$$

has a pole of order 3 at

$$z = 1.$$

Example 4.16. Consider

$$f(z) = \frac{z+1}{(z-2)^2(z+3)}.$$

At $z = 2$,

$$f(z) = \frac{1}{(z-2)^2} \frac{z+1}{z+3},$$

and

$$\frac{z+1}{z+3}$$

is holomorphic and nonzero at $z = 2$. Hence $z = 2$ is a pole of order 2.

Remark 4.17. Zeros and poles are dual notions:

$$f(z) = (z - z_0)^n g(z)$$

means that f has a zero of order n, whereas

$$f(z) = (z - z_0)^{-n} g(z)$$

means that f has a pole of order n, provided

$$g(z_0) \neq 0.$$

4.5 Laurent Series

Taylor series describe holomorphic functions in discs. If the center itself is a singular point, negative powers are required. This leads to Laurent series.

Definition 4.18 (Annulus)**.** An annulus centered at z_0 is a region of the form

$$A = \{z \in \mathbb{C} : r < |z - z_0| < R\},$$

where

$$0 \leq r < R \leq \infty.$$

Theorem 4.19 (Laurent Expansion)**.** *Let f be holomorphic in the annulus*

$$r < |z - z_0| < R.$$

Then f has a unique series representation

$$\boxed{f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n}$$

which converges in the annulus.

The coefficients are given by

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, \mathrm{d}\zeta,$$

where C is any positively oriented circle centered at z_0 lying inside the annulus.

The Laurent series naturally splits as

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}.$$

The first part is the ordinary Taylor part, while the second part records the singular behavior.

Definition 4.20 (Principal Part)**.** The principal part of the Laurent series at z_0 is

$$\sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}.$$

Remark 4.21. The principal part contains exactly the information describing the singularity.

Thus the three types of isolated singularities can be read directly from the negative-power terms of the Laurent series.

4.6 Examples of Laurent Expansions

A single rational function can have different Laurent expansions in different annuli.

Example 4.22. Consider

$$f(z) = \frac{1}{z(z-1)}.$$

For

$$0 < |z| < 1,$$

we write

$$\begin{aligned} f(z) &= -\frac{1}{z}\frac{1}{1-z} \\ &= -\frac{1}{z}\sum_{n=0}^{\infty} z^n \\ &= -\frac{1}{z} - 1 - z - z^2 - \cdots . \end{aligned}$$

For

$$|z| > 1,$$

we instead write

$$\begin{aligned} f(z) &= \frac{1}{z^2}\frac{1}{1-\frac{1}{z}} \\ &= \frac{1}{z^2}\sum_{n=0}^{\infty} \frac{1}{z^n} \\ &= \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \cdots . \end{aligned}$$

Remark 4.23. There is no single Laurent expansion valid everywhere.

The correct expansion depends on which ratio has modulus less than 1, because the geometric-series identity

$$\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n$$

requires

$$|w| < 1.$$

This is why the annulus of validity must always be stated together with a Laurent expansion.

Example 4.24. Consider

$$f(z) = \frac{1}{z-1}.$$

Centered at $z_0 = 0$, for

$$|z| < 1,$$

we have

$$f(z) = -\frac{1}{1-z} = -\sum_{n=0}^{\infty} z^n.$$

For

$$|z| > 1,$$

we have

$$\begin{aligned} f(z) &= \frac{1}{z}\frac{1}{1-\frac{1}{z}} \\ &= \sum_{n=1}^{\infty} \frac{1}{z^n}. \end{aligned}$$

4.7 Classification by Laurent Series

Laurent series give a complete classification of isolated singularities.

Theorem 4.25 (Classification of Isolated Singularities)**.** *Suppose f is holomorphic in*

$$0 < |z - z_0| < R$$

and has Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

Then:

1. *z_0 is removable if and only if*

$$a_{-1} = a_{-2} = a_{-3} = \cdots = 0.$$

2. *z_0 is a pole of order m if and only if*

$$a_{-m} \neq 0$$

and

$$a_{-n} = 0 \qquad \text{for all } n > m.$$

3. *z_0 is an essential singularity if and only if infinitely many negative-power coefficients are nonzero.*

Thus the three cases have the schematic forms

$$\begin{array}{|rcl|} \hline \text{removable} & : & a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots, \\ \text{pole} & : & \dfrac{a_{-m}}{(z-z_0)^m} + \cdots + \dfrac{a_{-1}}{z-z_0} + a_0 + \cdots, \\ \text{essential} & : & \cdots + \dfrac{a_{-2}}{(z-z_0)^2} + \dfrac{a_{-1}}{z-z_0} + a_0 + \cdots. \\ \hline \end{array}$$

4.8 Essential Singularities

Definition 4.26 (Essential Singularity)**.** An isolated singularity that is neither removable nor a pole is called an essential singularity.

Example 4.27. The function

$$f(z) = e^{1/z}$$

has the Laurent expansion

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \cdots.$$

There are infinitely many negative-power terms. Therefore

$$z = 0$$

is an essential singularity.

Example 4.28. The function

$$\sin \frac{1}{z}$$

has an essential singularity at $z = 0$, since

$$\sin \frac{1}{z} = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \cdots .$$

The behavior near an essential singularity is dramatically different from the behavior near a pole.

Theorem 4.29 (Casorati–Weierstrass Theorem)**.** *Suppose z_0 is an essential singularity of f. Then the image of every deleted neighborhood of z_0 under f is dense in $\mathbb{C}$.*

Equivalently, for every

$$w \in \mathbb{C},$$

there exists a sequence

$$z_n \to z_0$$

such that

$$f(z_n) \to w.$$

Proof. Suppose the conclusion is false. Then there exists

$$w_0 \in \mathbb{C}$$

and some $\varepsilon > 0$ such that

$$|f(z) - w_0| \geq \varepsilon$$

for all z sufficiently close to z_0.

Define

$$g(z) = \frac{1}{f(z) - w_0}.$$

Then g is holomorphic and bounded near z_0. By Riemann's theorem on removable singularities, g extends holomorphically to z_0.

If

$$g(z_0) \neq 0,$$

then

$$f(z) = w_0 + \frac{1}{g(z)}$$

also extends holomorphically to z_0, so the singularity is removable.

If

$$g(z_0) = 0,$$

then g has a zero of some finite order m, so

$$\frac{1}{g(z)}$$

has a pole of order m. Hence f has a pole at z_0.

Both possibilities contradict the assumption that z_0 is an essential singularity. Therefore the image is dense in $\mathbb{C}$. □

Remark 4.30. The three types of isolated singularities have sharply different behavior:

removable	$\Longrightarrow$	$f(z)$ remains bounded and approaches a finite value,
pole	$\Longrightarrow$	$\lvert f(z)\rvert \to \infty$,
essential	$\Longrightarrow$	$f(z)$ comes arbitrarily close to every complex value.

4.9 Meromorphic Functions

Definition 4.31 (Meromorphic Function)**.** A function f is meromorphic in an open set Ω if there exists a set of isolated points

$$\{z_1, z_2, \ldots\} \subset \Omega$$

having no limit point in Ω, such that

1. f is holomorphic in
$$\Omega \setminus \{z_1, z_2, \ldots\},$$
2. each z_j is a pole of f.

Thus a meromorphic function is holomorphic except for isolated poles.

Example 4.32. Every rational function

$$R(z) = \frac{P(z)}{Q(z)}$$

is meromorphic in $\mathbb{C}$. Its poles occur among the zeros of Q, after any common factors of P and Q are cancelled.

Remark 4.33. An essential singularity is not allowed for a meromorphic function.

Thus

$$e^{1/z}$$

is holomorphic on

$$\mathbb{C} \setminus \{0\},$$

but it is not meromorphic on all of $\mathbb{C}$, because 0 is an essential singularity rather than a pole.

4.10 The Point at Infinity

It is often useful to enlarge the complex plane by adding one point, denoted by

$$\infty.$$

Definition 4.34 (Extended Complex Plane)**.** The extended complex plane is

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

It is also called the Riemann sphere.

The geometry of the Riemann sphere can be understood through stereographic projection. Under this correspondence, the north pole of the sphere represents

$$\infty.$$

Remark 4.35. The Riemann sphere turns "behavior as $|z| \to \infty$" into ordinary local behavior near a point.

This allows infinity to be studied using exactly the same language as finite isolated singularities.

Definition 4.36 (Singularity at Infinity)**.** Let f be defined for sufficiently large $|z|$. Define

$$F(w) = f\left(\frac{1}{w}\right).$$

The nature of the singularity of f at infinity is defined to be the nature of the singularity of F at

$$w = 0.$$

Thus:

$$\begin{aligned} f \text{ has a removable singularity at } \infty &\iff f(1/w) \text{ has a removable singularity at } 0, \\ f \text{ has a pole at } \infty &\iff f(1/w) \text{ has a pole at } 0, \\ f \text{ has an essential singularity at } \infty &\iff f(1/w) \text{ has an essential singularity at } 0. \end{aligned}$$

Example 4.37. Let

$$f(z) = z^n.$$

Then

$$f(1/w) = \frac{1}{w^n},$$

so f has a pole of order n at infinity.

Example 4.38. For

$$f(z) = e^z,$$

we have

$$f(1/w) = e^{1/w}.$$

Hence e^z has an essential singularity at infinity.

Example 4.39. For

$$f(z) = \frac{1}{z},$$

we have

$$f(1/w) = w.$$

Thus infinity is a removable singularity, and the natural extension is

$$f(\infty) = 0.$$

4.11 Meromorphic Functions on the Riemann Sphere

Theorem 4.40. *Every meromorphic function on the extended complex plane*

$$\widehat{\mathbb{C}}$$

is a rational function.

Remark 4.41. This result gives a useful global characterization:

$$\boxed{\text{meromorphic on the Riemann sphere} \iff \text{rational.}}$$

Thus rational functions are precisely the globally meromorphic functions when the complex plane is completed by the point at infinity.

4.12 A Practical Classification Procedure

Suppose z_0 is an isolated singularity of f. A useful procedure is:

1. First check whether
$$\lim_{z \to z_0} f(z)$$
exists and is finite. If so, the singularity is removable.
2. If not, check whether
$$|f(z)| \to \infty.$$
If so, z_0 is a pole.
3. More precisely, find the smallest $m \geq 1$ such that
$$(z - z_0)^m f(z)$$
has a finite nonzero limit at z_0. Then z_0 is a pole of order m.
4. If neither case occurs, examine the Laurent expansion. Infinitely many nonzero negative-power terms imply an essential singularity.

Example 4.42. Consider
$$f(z) = \frac{1 - \cos z}{z^2}.$$
Since
$$1 - \cos z = \frac{z^2}{2} - \frac{z^4}{4!} + \cdots,$$
we obtain
$$f(z) = \frac{1}{2} - \frac{z^2}{4!} + \cdots.$$
There are no negative powers, so $z = 0$ is removable.

Example 4.43. Consider
$$f(z) = \frac{\sin z}{z^4}.$$
Since
$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots,$$
we obtain
$$f(z) = \frac{1}{z^3} - \frac{1}{3!z} + \frac{z}{5!} - \cdots.$$
The principal part has finitely many terms, with highest negative power
$$z^{-3}.$$
Therefore $z = 0$ is a pole of order 3.

Example 4.44. Consider
$$f(z) = e^{1/z}.$$
Its Laurent expansion contains infinitely many negative powers:
$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \cdots.$$
Therefore $z = 0$ is an essential singularity.

4.13 Main Structure of the Chapter

The central idea of this chapter is that isolated singularities are not arbitrary. Laurent series completely encode their behavior:

$$
\boxed{
\begin{array}{c}
f \text{ holomorphic on } 0 < |z - z_0| < R \\
\Downarrow \\
f(z) = \displaystyle\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \\
\Downarrow \\
\text{inspect the principal part} \\
\Downarrow \\
\begin{cases}
\text{no negative powers} & \Rightarrow \text{removable}, \\
\text{finitely many negative powers} & \Rightarrow \text{pole}, \\
\text{infinitely many negative powers} & \Rightarrow \text{essential}.
\end{cases}
\end{array}
}
$$

The next chapter will show that among all the coefficients in the principal part, one coefficient has a particularly important role:

$$a_{-1}.$$

It is the residue, and it converts the local singular behavior of a function into the value of a global contour integral.

Chapter 5

Residues and the Argument Principle

5 Residues and the Argument Principle

5.1 Residues

Let f have an isolated singularity at z_0, with Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

Among all coefficients in the Laurent expansion, the coefficient of

$$\frac{1}{z - z_0}$$

plays a distinguished role in contour integration.

Definition 5.1 (Residue)**.** The residue of f at z_0 is defined by

$$\boxed{\operatorname{Res}_{z=z_0} f = a_{-1}.}$$

Remark 5.2. The residue is only one coefficient in the Laurent expansion, but it is the only coefficient that contributes to a contour integral around the singularity.

Indeed,

$$\int_{|z-z_0|=r} (z - z_0)^n \, \mathrm{d}z = 0$$

for every integer $n \neq -1$, while

$$\int_{|z-z_0|=r} \frac{1}{z - z_0} \, \mathrm{d}z = 2\pi i.$$

Theorem 5.3. *Let $P(z)$ be the principal part*

$$P(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + \frac{a_{-1}}{z - z_0}.$$

Then

$$\frac{1}{2\pi i} \int_C P(z) \, \mathrm{d}z = a_{-1},$$

where C is any positively oriented circle centered at z_0.

Proof. By linearity,

$$\int_C P(z) \, \mathrm{d}z = \sum_{k=1}^{m} a_{-k} \int_C \frac{\mathrm{d}z}{(z - z_0)^k}.$$

All terms vanish except the term $k = 1$, for which

$$\int_C \frac{\mathrm{d}z}{z - z_0} = 2\pi i.$$

Hence

$$\int_C P(z) \, \mathrm{d}z = 2\pi i \, a_{-1}.$$

□

5.2 Residues at Poles

Residues can often be computed directly without writing the entire Laurent series.

Theorem 5.4 (Residue at a Simple Pole)**.** *If f has a simple pole at z_0, then*

$$\boxed{\operatorname{Res}_{z=z_0} f = \lim_{z\to z_0}(z-z_0)f(z).}$$

Proof. Since the pole is simple,

$$f(z) = \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \cdots .$$

Multiplying by $z - z_0$,

$$(z-z_0)f(z) = a_{-1} + a_0(z-z_0) + \cdots .$$

Letting $z \to z_0$ gives

$$a_{-1} = \operatorname{Res}_{z=z_0} f.$$

□

Corollary 5.5. *Suppose*

$$f(z) = \frac{g(z)}{h(z)},$$

where g and h are holomorphic near z_0,

$$h(z_0) = 0, \qquad h'(z_0) \neq 0,$$

and

$$g(z_0) \neq 0.$$

Then z_0 is a simple pole and

$$\boxed{\operatorname{Res}_{z=z_0} f = \frac{g(z_0)}{h'(z_0)}.}$$

Proof. Since

$$h(z) = h'(z_0)(z-z_0) + o(z-z_0),$$

we have

$$\begin{aligned}\operatorname{Res}_{z=z_0} f &= \lim_{z\to z_0}(z-z_0)\frac{g(z)}{h(z)} \\ &= \frac{g(z_0)}{h'(z_0)}.\end{aligned}$$

□

Example 5.6. Consider

$$f(z) = \frac{e^z}{z^2+1}.$$

The poles are

$$z = i, \qquad z = -i.$$

At $z = i$,

$$\operatorname{Res}_{z=i} f = \frac{e^i}{2i}.$$

Similarly,

$$\operatorname{Res}_{z=-i} f = \frac{e^{-i}}{-2i}.$$

Theorem 5.7 (Residue at a Pole of Order m)**.** *Suppose f has a pole of order m at z_0. Then*

$$\boxed{\operatorname{Res}_{z=z_0} f = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}} \left[(z - z_0)^m f(z) \right].}$$

Proof. Write

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + \frac{a_{-1}}{z - z_0} + a_0 + \cdots .$$

Then

$$\begin{aligned}(z - z_0)^m f(z) = a_{-m} + a_{-(m-1)}(z - z_0) + \cdots \\ + a_{-1}(z - z_0)^{m-1} + \cdots .\end{aligned}$$

Differentiating $m - 1$ times and evaluating at z_0, all terms vanish except

$$(m-1)! a_{-1}.$$

Hence

$$a_{-1} = \frac{1}{(m-1)!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}} \Big[(z - z_0)^m f(z) \Big] \Big|_{z=z_0}.$$

□

Example 5.8. Let

$$f(z) = \frac{e^z}{z^3}.$$

Then $z = 0$ is a pole of order 3. Therefore

$$\begin{aligned}\operatorname{Res}_{z=0} f &= \frac{1}{2!} \frac{\mathrm{d}^2}{\mathrm{d}z^2} e^z \Big|_{z=0} \\ &= \frac{1}{2}.\end{aligned}$$

5.3 The Residue Theorem

The residue theorem is one of the central results of complex analysis. It reduces a global contour integral to local information at isolated singularities.

Theorem 5.9 (Residue Theorem)**.** *Let f be holomorphic in a region Ω except at finitely many isolated singularities*

$$z_1, \ldots, z_n.$$

Let γ be a positively oriented simple closed curve in Ω which contains all $z_1, \ldots, z_n$ in its interior and does not pass through any singularity. Then

$$\boxed{\int_\gamma f(z)\, \mathrm{d}z = 2\pi i \sum_{k=1}^{n} \operatorname{Res}_{z=z_k} f.}$$

Proof. Choose pairwise disjoint small positively oriented circles

$$C_1, \ldots, C_n$$

around the singularities $z_1, \ldots, z_n$, all contained inside γ.

The function f is holomorphic in the region bounded by γ and the circles C_k. Hence Cauchy's theorem gives

$$\int_\gamma f(z)\ \mathrm{d}z = \sum_{k=1}^{n} \int_{C_k} f(z)\ \mathrm{d}z.$$

For each k,

$$\int_{C_k} f(z)\ \mathrm{d}z = 2\pi i \operatorname{Res}_{z=z_k} f.$$

Summing over all singularities gives the result. □

Remark 5.10. The residue theorem has the striking form

$$\boxed{\text{global contour integral} = \text{sum of local singular data.}}$$

The shape of the contour is largely irrelevant. What matters is which singularities lie inside it and what their residues are.

Example 5.11. Evaluate

$$\int_{|z|=2} \frac{1}{z(z-1)}\ \mathrm{d}z.$$

The integrand has simple poles at

$$z = 0, \qquad z = 1.$$

Their residues are

$$\operatorname{Res}_{z=0} \frac{1}{z(z-1)} = -1,$$

and

$$\operatorname{Res}_{z=1} \frac{1}{z(z-1)} = 1.$$

Therefore

$$\int_{|z|=2} \frac{1}{z(z-1)}\ \mathrm{d}z = 2\pi i(-1+1) = 0.$$

5.4 A Practical Procedure for Residue Calculations

For contour-integral problems, the following procedure is useful:

1. Find all singularities of f.
2. Determine which singularities lie inside the contour.
3. Determine the order of each pole.
4. Compute each residue, using one of:

   $$\operatorname{Res}_{z=z_0} f = \lim_{z\to z_0} (z-z_0) f(z)$$

   for a simple pole,
   or

   $$\operatorname{Res}_{z=z_0} f = \frac{1}{(m-1)!} \lim_{z\to z_0} \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}} \left[(z-z_0)^m f(z)\right]$$

   for a pole of order m.
5. Apply

   $$\int_\gamma f(z)\ \mathrm{d}z = 2\pi i \sum \operatorname{Res} f.$$

Remark 5.12. When a Laurent expansion is easy to obtain, the residue can also be read directly as the coefficient of

$$(z-z_0)^{-1}.$$

One should use whichever method produces the coefficient most efficiently.

5.5 Trigonometric Integrals

Residues can be used to evaluate integrals of the form

$$\int_0^{2\pi} F(\cos\theta, \sin\theta)\ \mathrm{d}\theta.$$

The standard substitution is

$$z = e^{i\theta}, \qquad |z| = 1.$$

Then

$$\mathrm{d}z = iz\ \mathrm{d}\theta,$$

so

$$\mathrm{d}\theta = \frac{\mathrm{d}z}{iz}.$$

Moreover,

$$\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right),$$

and

$$\sin\theta = \frac{1}{2i}\left(z - \frac{1}{z}\right).$$

Therefore

$$\boxed{\begin{aligned}&\int_0^{2\pi} F(\cos\theta, \sin\theta)\ \mathrm{d}\theta \\ &\qquad = \int_{|z|=1} F\left(\frac{1}{2}\left(z + \frac{1}{z}\right), \frac{1}{2i}\left(z - \frac{1}{z}\right)\right)\frac{\mathrm{d}z}{iz}.\end{aligned}}$$

Example 5.13. Evaluate

$$I = \int_0^{2\pi} \frac{\mathrm{d}\theta}{a + b\cos\theta},$$

where

$$a > |b| > 0.$$

Using

$$z = e^{i\theta},$$

we obtain

$$\begin{aligned} I &= \int_{|z|=1} \frac{1}{a + \frac{b}{2}(z + z^{-1})}\frac{\mathrm{d}z}{iz} \\ &= \frac{2}{i}\int_{|z|=1} \frac{\mathrm{d}z}{bz^2 + 2az + b}. \end{aligned}$$

The roots are

$$z_\pm = \frac{-a \pm \sqrt{a^2 - b^2}}{b}.$$

Since

$$z_+ z_- = 1,$$

exactly one root lies inside the unit circle. Applying the residue theorem gives

$$\boxed{I = \frac{2\pi}{\sqrt{a^2 - b^2}}.}$$

5.6 Real Integrals over the Real Line

Residues can also evaluate improper integrals of the form

$$\int_{-\infty}^{\infty} f(x)\ \mathrm{d}x.$$

A standard contour is the semicircle

$$C_R = [-R, R] \cup \Gamma_R,$$

where

$$\Gamma_R = \{Re^{it} : 0 \leq t \leq \pi\}.$$

Then

$$\int_{C_R} f(z)\ \mathrm{d}z = \int_{-R}^{R} f(x)\ \mathrm{d}x + \int_{\Gamma_R} f(z)\ \mathrm{d}z.$$

If

$$\int_{\Gamma_R} f(z)\ \mathrm{d}z \longrightarrow 0$$

as $R \to \infty$, the residue theorem gives

$$\boxed{\int_{-\infty}^{\infty} f(x)\ \mathrm{d}x = 2\pi i \sum_{\Im z_k > 0} \mathrm{Res}_{z=z_k} f.}$$

Proposition 5.14. *Let*

$$f(z) = \frac{P(z)}{Q(z)},$$

where

$$\deg Q \geq \deg P + 2.$$

Then on the upper semicircle

$$\Gamma_R = \{Re^{it} : 0 \leq t \leq \pi\},$$

we have

$$\int_{\Gamma_R} f(z)\ \mathrm{d}z \longrightarrow 0$$

as

$$R \to \infty.$$

Proof. For sufficiently large R,

$$|f(z)| \leq \frac{C}{R^2}$$

on Γ_R. Since

$$L(\Gamma_R) = \pi R,$$

the ML-estimate gives

$$\left|\int_{\Gamma_R} f(z)\ \mathrm{d}z\right| \leq \frac{C}{R^2}\pi R = \frac{\pi C}{R} \longrightarrow 0.$$

□

Example 5.15. Evaluate

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2+1}.$$

Consider

$$f(z) = \frac{1}{z^2+1}.$$

In the upper half-plane there is one pole,

$$z = i.$$

Its residue is

$$\mathrm{Res}_{z=i}\, f = \frac{1}{2i}.$$

The semicircular integral tends to zero, so

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2+1} &= 2\pi i \frac{1}{2i} \\ &= \pi.\end{aligned}$$

5.7 Fourier-Type Integrals

To evaluate integrals involving

$$\cos(ax) \qquad \text{or} \qquad \sin(ax),$$

it is often useful to consider

$$e^{iaz}$$

because

$$e^{iax} = \cos(ax) + i\sin(ax).$$

Thus

$$\begin{aligned}\int_{-\infty}^{\infty} f(x)e^{iax}\ \mathrm{d}x = &\int_{-\infty}^{\infty} f(x)\cos(ax)\ \mathrm{d}x \\ &+ i\int_{-\infty}^{\infty} f(x)\sin(ax)\ \mathrm{d}x.\end{aligned}$$

The real and imaginary parts then give the desired cosine and sine integrals.

Example 5.16. For $a > 0$,

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{x^2+1}\ \mathrm{d}x$$

is evaluated using the upper half-plane, because

$$e^{iaz} = e^{iax-ay}$$

decays when

$$y > 0.$$

The only pole in the upper half-plane is $z = i$. Hence

$$\begin{aligned}\mathrm{Res}_{z=i}\, \frac{e^{iaz}}{z^2+1} &= \frac{e^{iai}}{2i} \\ &= \frac{e^{-a}}{2i}.\end{aligned}$$

Therefore

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{x^2+1}\,\mathrm{d}x = \pi e^{-a}.$$

Taking real parts gives

$$\boxed{\int_{-\infty}^{\infty} \frac{\cos(ax)}{x^2+1}\,\mathrm{d}x = \pi e^{-a}.}$$

Remark 5.17. The sign of a determines which half-plane should be used.

For

$$a > 0,$$

the factor

$$e^{iaz} = e^{iax-ay}$$

decays in the upper half-plane.

For

$$a < 0,$$

it decays in the lower half-plane.

5.8 Poles on the Real Axis

Sometimes the integrand has a pole directly on the real axis. The contour must then be modified by a small semicircular indentation around the pole.

Suppose f has a simple pole at

$$z = c \in \mathbb{R}.$$

Let

$$C_\varepsilon : z = c + \varepsilon e^{it}, \qquad \pi \geq t \geq 0,$$

be a small semicircle in the upper half-plane traversed clockwise.

Then

$$\boxed{\lim_{\varepsilon\to 0} \int_{C_\varepsilon} f(z)\,\mathrm{d}z = -i\pi \operatorname{Res}_{z=c} f.}$$

If the small semicircle is traversed counterclockwise, the sign is reversed.

Remark 5.18. This is the origin of the "half-residue" contribution that appears in principal-value contour integrals:

$$\pi i \operatorname{Res}_{z=c} f$$

instead of

$$2\pi i \operatorname{Res}_{z=c} f.$$

5.9 The Logarithmic Derivative

Let f be meromorphic. The function

$$\frac{f'(z)}{f(z)}$$

encodes the zeros and poles of f.

Suppose f has a zero of order m at z_0. Then

$$f(z) = (z-z_0)^m g(z), \qquad g(z_0) \neq 0.$$

Hence

$$\frac{f'(z)}{f(z)} = \frac{m}{z - z_0} + \frac{g'(z)}{g(z)}.$$

Therefore

$$\operatorname{Res}_{z=z_0} \frac{f'}{f} = m.$$

Similarly, if f has a pole of order m at z_0, then

$$\operatorname{Res}_{z=z_0} \frac{f'}{f} = -m.$$

Thus

$$\boxed{\operatorname{Res}\left(\frac{f'}{f}\right) = \begin{cases} +m, & \text{zero of order } m, \\ -m, & \text{pole of order } m. \end{cases}}$$

5.10 The Argument Principle

Theorem 5.19 (Argument Principle)**.** *Let f be meromorphic inside and on a positively oriented simple closed curve C. Assume that f has no zeros or poles on C.*

Let

$$N$$

be the number of zeros of f inside C, counted with multiplicity, and let

$$P$$

be the number of poles inside C, also counted with multiplicity.

Then

$$\boxed{\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} \, \mathrm{d}z = N - P.}$$

Proof. The only singularities of

$$\frac{f'}{f}$$

inside C occur at zeros and poles of f.

At a zero of multiplicity m, the residue is m. At a pole of order m, the residue is $-m$.

Therefore, by the residue theorem,

$$\begin{aligned} \frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} \, \mathrm{d}z &= \sum_{\text{zeros}} m - \sum_{\text{poles}} m \\ &= N - P. \end{aligned}$$

□

Remark 5.20. The argument principle transforms analytic information into a counting problem:

$$\boxed{\text{boundary integral} \Longrightarrow \text{number of zeros minus poles inside.}}$$

This is another local-to-global principle: the behavior of f along the boundary determines discrete information about the interior.

5.11 Geometric Meaning of the Argument Principle

Along the curve C,

$$f(z)$$

traces another closed curve in the complex plane.

Since formally

$$\mathrm{d}\log f(z) = \frac{f'(z)}{f(z)}\,\mathrm{d}z,$$

we have

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,\mathrm{d}z$$

equal to the net number of times the curve $f(C)$ winds around the origin.

Thus

$$N - P$$

is the winding number of $f(C)$ around 0.

Remark 5.21. This explains the name "argument principle."

As z moves once around C, the total change in

$$\arg f(z)$$

is

$$2\pi(N - P).$$

5.12 Rouché's Theorem

The argument principle leads to a powerful theorem for counting zeros without solving an equation explicitly.

Theorem 5.22 (Rouché's Theorem)**.** *Let f and g be holomorphic inside and on a simple closed curve C. Suppose*

$$|g(z)| < |f(z)|$$

for every

$$z \in C.$$

Then

$$f \qquad \text{and} \qquad f + g$$

have the same number of zeros inside C, counted with multiplicity.

Proof. For

$$0 \le t \le 1,$$

define

$$F_t(z) = f(z) + tg(z).$$

On C,

$$|tg(z)| \le |g(z)| < |f(z)|.$$

Hence

$$F_t(z) \neq 0$$

on C for every $t \in [0, 1]$.

The number of zeros inside C is given by

$$N(t) = \frac{1}{2\pi i} \int_C \frac{F_t'(z)}{F_t(z)} \, \mathrm{d}z.$$

As t varies continuously, $N(t)$ varies continuously. But $N(t)$ is integer-valued. Therefore $N(t)$ must be constant.

Thus

$$N(0) = N(1).$$

Since

$$F_0 = f, \qquad F_1 = f + g,$$

the two functions have the same number of zeros inside C. □

Remark 5.23. The intuitive idea is that if

$$|g| < |f|$$

on the boundary, then adding g to f is too small a perturbation to change how many times the boundary image winds around 0.

Hence the number of interior zeros cannot change.

Example 5.24. Determine the number of zeros of

$$P(z) = z^5 + 4z + 1$$

inside

$$|z| < 2.$$

On

$$|z| = 2,$$

we have

$$|z^5| = 32,$$

while

$$|4z + 1| \leq 8 + 1 = 9.$$

Therefore

$$|4z + 1| < |z^5|.$$

By Rouché's theorem,

$$z^5$$

and

$$z^5 + 4z + 1$$

have the same number of zeros inside $|z| < 2$.

Since z^5 has five zeros at the origin counted with multiplicity, P has exactly

$$5$$

zeros inside the disc.

Example 5.25. Determine the number of zeros of

$$z^4 + z + 1$$

inside the unit disc.

On

$$|z| = 1,$$

the naive choice

$$f(z) = z^4$$

does not work, because

$$|z + 1|$$

is not always strictly smaller than 1.

This illustrates an important point: Rouché's theorem is powerful, but the choice of dominant term must be adapted to the contour. A failed inequality does not imply the theorem is useless; it only means that a different decomposition or contour is required.

5.13 Residues and Counting Zeros

The residue theorem and the argument principle can now be viewed as two forms of the same mechanism.

For a general meromorphic function,

$$\int_C f(z)\, \mathrm{d}z$$

extracts the coefficients

$$a_{-1}$$

at its singularities.

For the logarithmic derivative,

$$\frac{f'}{f},$$

these coefficients are integers representing multiplicities.

Thus

Residue theorem:	$\dfrac{1}{2\pi i}\displaystyle\int_C f(z)\, \mathrm{d}z = \sum \operatorname{Res} f,$
Argument principle:	$\dfrac{1}{2\pi i}\displaystyle\int_C \frac{f'}{f}\, \mathrm{d}z = N - P.$

5.14 Main Structure of the Chapter

The logical structure of this chapter is

Laurent expansion

$\Downarrow$

$a_{-1} = \operatorname{Res}_{z=z_0} f$

$\Downarrow$

Residue Theorem

$\Downarrow$

evaluation of contour and real integrals.

At the same time,

$$\boxed{\begin{array}{c}\text{zeros and poles of } f\\ \Downarrow\\ \operatorname{Res}\left(\dfrac{f'}{f}\right) = \text{multiplicity}\\ \Downarrow\\ \text{Argument Principle}\\ \Downarrow\\ \text{Rouché's Theorem}\\ \Downarrow\\ \text{counting zeros without solving equations.}\end{array}}$$

The residue theorem therefore has two complementary roles in complex analysis:

$$\boxed{\text{it evaluates integrals} \qquad \text{and} \qquad \text{it counts zeros and poles.}}$$

Chapter 6

The Complex Logarithm and Holomorphic Branches

6 The Complex Logarithm and Holomorphic Branches

6.1 Argument of a Complex Number

Every nonzero complex number can be written in polar form as

$$z = re^{i\theta}, \qquad r = |z| > 0.$$

Unlike the modulus r, the angle θ is not uniquely determined.

Definition 6.1 (Argument)**.** Let $z \in \mathbb{C} \setminus \{0\}$. A real number θ is called an argument of z if

$$z = |z|e^{i\theta}.$$

If θ is one argument of z, then all arguments are

$$\theta + 2\pi k, \qquad k \in \mathbb{Z}.$$

We write

$$\arg z = \{\theta + 2\pi k : k \in \mathbb{Z}\}.$$

Example 6.2. For

$$z = i,$$

the arguments are

$$\arg i = \left\{\frac{\pi}{2} + 2\pi k : k \in \mathbb{Z}\right\}.$$

Remark 6.3. The multivalued nature of the argument comes from the periodicity

$$e^{i(\theta+2\pi k)} = e^{i\theta}.$$

Geometrically, rotating once around the origin changes the continuously chosen angle by 2π, even though one returns to the same point in the complex plane. This is the basic topological obstruction behind the complex logarithm.

6.2 The Complex Logarithm

For real $x > 0$, the logarithm is the inverse of the exponential function. In the complex plane, however, the exponential function is periodic:

$$e^{z+2\pi i} = e^z.$$

Therefore it cannot have a single-valued inverse on all of $\mathbb{C} \setminus \{0\}$.

Definition 6.4 (Complex Logarithm)**.** Let

$$z = re^{i\theta}, \qquad r > 0.$$

The complex logarithm of z is the multivalued expression

$$\boxed{\log z = \log|z| + i \arg z.}$$

Equivalently,

$$\log z = \{\log|z| + i(\theta + 2\pi k) : k \in \mathbb{Z}\}.$$

Thus if $w \in \log z$, then

$$e^w = z.$$

Conversely, if

$$e^w = z,$$

then

$$w \in \log z.$$

Example 6.5. Since

$$1 = e^{2\pi i k}, \qquad k \in \mathbb{Z},$$

we have

$$\log 1 = \{2\pi i k : k \in \mathbb{Z}\}.$$

In particular, the complex logarithm of 1 is not just 0.

Example 6.6. For

$$z = -1,$$

we have

$$\arg(-1) = \{(2k+1)\pi : k \in \mathbb{Z}\}.$$

Hence

$$\log(-1) = \{(2k+1)\pi i : k \in \mathbb{Z}\}.$$

Remark 6.7. One must distinguish carefully between

$$\log z$$

as a multivalued expression and a particular single-valued branch of the logarithm.

The difficulty is therefore not defining possible logarithmic values at one point. The difficulty is choosing one of those values continuously, and preferably holomorphically, throughout an entire domain.

6.3 Branches of the Argument and Logarithm

Definition 6.8 (Branch of the Argument)**.** Let

$$\Omega \subset \mathbb{C} \setminus \{0\}$$

be a domain. A continuous function

$$\Theta : \Omega \to \mathbb{R}$$

is called a branch of the argument if

$$z = |z| e^{i\Theta(z)}$$

for every $z \in \Omega$.

Once a continuous branch of the argument is chosen, a single-valued logarithm can be defined.

Definition 6.9 (Branch of the Logarithm)**.** If Θ is a branch of the argument on Ω, define

$$L(z) = \log|z| + i\Theta(z).$$

Then

$$e^{L(z)} = z.$$

Such a function is called a branch of the logarithm.

Proposition 6.10. *Every holomorphic branch L of the logarithm satisfies*

$$\boxed{L'(z) = \frac{1}{z}.}$$

Proof. Since

$$e^{L(z)} = z,$$

differentiate both sides:

$$e^{L(z)}L'(z) = 1.$$

Since

$$e^{L(z)} = z,$$

we obtain

$$zL'(z) = 1,$$

and therefore

$$L'(z) = \frac{1}{z}.$$

□

This immediately connects the logarithm with the theory of primitives from Chapter 2:

$$\boxed{\text{a holomorphic logarithm of } z \quad \Longrightarrow \quad \frac{1}{z} \text{ has a primitive.}}$$

In fact, the converse is also true.

Theorem 6.11 (Logarithm and Primitive of $1/z$)**.** *Let*

$$\Omega \subset \mathbb{C} \setminus \{0\}$$

be a region. Then the following are equivalent:

1. *there exists a holomorphic branch of* $\log z$ *on* Ω*;*
2. *the function*

$$\frac{1}{z}$$

has a primitive on Ω*.*

Proof. The implication

$$(1) \Longrightarrow (2)$$

follows from

$$L'(z) = \frac{1}{z}.$$

Conversely, suppose F is a primitive of $1/z$:

$$F'(z) = \frac{1}{z}.$$

Consider

$$h(z) = ze^{-F(z)}.$$

Then

$$\begin{aligned} h'(z) &= e^{-F(z)} - ze^{-F(z)}F'(z) \\ &= e^{-F(z)} - ze^{-F(z)}\frac{1}{z} \\ &= 0. \end{aligned}$$

Since Ω is connected, h is constant:

$$ze^{-F(z)} = c$$

for some $c \neq 0$.

Choose $a \in \mathbb{C}$ such that

$$e^a = c.$$

Then

$$e^{F(z)+a} = z.$$

Therefore

$$L(z) = F(z) + a$$

is a holomorphic branch of $\log z$. □

6.4 The Principal Branch

A global branch of the logarithm cannot be defined on

$$\mathbb{C} \setminus \{0\},$$

but it can be defined after removing a suitable ray from the origin.

Definition 6.12 (Principal Argument)**.** For

$$z \in \mathbb{C} \setminus (-\infty, 0],$$

the principal argument is the unique argument satisfying

$$-\pi < z < \pi.$$

Definition 6.13 (Principal Branch of the Logarithm)**.** The principal branch of the logarithm is

$$\boxed{z = \log|z| + iz,}$$

defined on

$$\mathbb{C} \setminus (-\infty, 0].$$

It satisfies

$$e^z = z$$

and

$$(z)' = \frac{1}{z}.$$

Example 6.14. For $z = i$,

$$i = \frac{\pi}{2},$$

so

$$i = \frac{\pi i}{2}.$$

For $z = -i$,

$$(-i) = -\frac{\pi}{2},$$

so

$$(-i) = -\frac{\pi i}{2}.$$

Remark 6.15. The negative real axis is removed because the argument jumps there: approaching a negative real number from above gives

$$z \to \pi,$$

while approaching it from below gives

$$z \to -\pi.$$

The difference is

$$2\pi.$$

Thus the branch cut is not a singularity of $1/z$ itself. It is introduced to make a single-valued continuous choice of argument possible.

Remark 6.16. The principal branch is only one possible branch. For example, one could instead choose

$$0 < z < 2\pi,$$

which produces a logarithm with branch cut along the positive real axis.

There is no mathematically preferred branch in every problem; the principal branch is simply the standard convention.

6.5 Why There Is No Global Logarithm on the Punctured Plane

The function

$$\frac{1}{z}$$

is holomorphic on

$$\mathbb{C} \setminus \{0\}.$$

Nevertheless, it has no primitive there.

As shown in Chapter 2,

$$\int_{|z|=1} \frac{\mathrm{d}z}{z} = 2\pi i.$$

If a holomorphic logarithm L existed on

$$\mathbb{C} \setminus \{0\},$$

then

$$L'(z) = \frac{1}{z}.$$

The fundamental theorem for contour integrals would then imply

$$\int_{|z|=1} \frac{\mathrm{d}z}{z} = 0,$$

which contradicts

$$\int_{|z|=1} \frac{\mathrm{d}z}{z} = 2\pi i.$$

Therefore:

$$\boxed{\text{there is no holomorphic logarithm on } \mathbb{C} \setminus \{0\}.}$$

Remark 6.17. This failure is topological rather than local.

Near every point

$$z_0 \neq 0,$$

a local branch of the logarithm exists. The obstruction appears only when one attempts to continue the branch around a closed curve that winds around the origin.

6.6 Simply Connected Domains and the Logarithm

We proved in Chapter 2 that every holomorphic function on a simply connected domain has a primitive. Applying that result to $1/z$ immediately gives the fundamental existence theorem for logarithms.

Theorem 6.18 (Existence of a Holomorphic Logarithm)**.** *Let*

$$\Omega \subset \mathbb{C} \setminus \{0\}$$

be a simply connected domain. Then there exists a holomorphic function

$$L : \Omega \to \mathbb{C}$$

such that

$$e^{L(z)} = z.$$

Equivalently, Ω admits a holomorphic branch of the logarithm.

Proof. Since

$$0 \notin \Omega,$$

the function

$$f(z) = \frac{1}{z}$$

is holomorphic on Ω.

By the primitive theorem for simply connected domains proved in Chapter 2, there exists F such that

$$F'(z) = \frac{1}{z}.$$

By the equivalence established above, F, after addition of a suitable constant, determines a holomorphic branch of the logarithm. □

Remark 6.19. This theorem explains the role of simply connectedness:

$$\boxed{\begin{array}{c}\Omega \text{ simply connected}, \qquad 0 \notin \Omega \\ \Downarrow \\ \dfrac{1}{z} \text{ has a primitive} \\ \Downarrow \\ \log z \text{ has a holomorphic branch on } \Omega.\end{array}}$$

The topology of the domain therefore determines whether the multivalued logarithm can be made single-valued.

Example 6.20. The right half-plane

$$\Omega = \{z : \Re z > 0\}$$

is simply connected and does not contain 0. Hence it admits a holomorphic logarithm.

One convenient choice of argument is

$$-\frac{\pi}{2} < z < \frac{\pi}{2}.$$

Example 6.21. The punctured disc

$$0 < |z| < 1$$

does not admit a global holomorphic logarithm.

Indeed, the unit circle of sufficiently small radius lies in the domain and winds around 0, and

$$\int_{|z|=r} \frac{\mathrm{d}z}{z} = 2\pi i.$$

6.7 Holomorphic Logarithms of Nonvanishing Functions

The preceding discussion concerns the logarithm of the coordinate function z. More generally, one can ask when a nonvanishing holomorphic function f can be written as an exponential.

Definition 6.22 (Holomorphic Logarithm of a Function)**.** Let f be holomorphic and nonvanishing in a domain Ω. A holomorphic function g satisfying

$$e^{g(z)} = f(z)$$

is called a holomorphic logarithm of f.

If such a g exists, differentiating

$$e^g = f$$

gives

$$e^g g' = f',$$

and hence

$$\boxed{g'(z) = \frac{f'(z)}{f(z)}.}$$

Thus the logarithmic derivative from Chapter 5 appears naturally.

Theorem 6.23 (Existence of a Logarithm of a Nonvanishing Function)**.** *Let Ω be simply connected, and let f be holomorphic and nonvanishing on Ω. Then f has a holomorphic logarithm on Ω.*

Proof. Since f never vanishes,

$$\frac{f'}{f}$$

is holomorphic on Ω.

Because Ω is simply connected, it has a primitive. Thus there exists a holomorphic function G satisfying

$$G'(z) = \frac{f'(z)}{f(z)}.$$

Consider

$$H(z) = f(z)e^{-G(z)}.$$

Then

$$\begin{aligned} H'(z) &= f'(z)e^{-G(z)} - f(z)e^{-G(z)}G'(z) \\ &= e^{-G(z)}\left[f'(z) - f(z)\frac{f'(z)}{f(z)}\right] \\ &= 0. \end{aligned}$$

Hence

$$H(z) = c$$

for some nonzero constant c.

Choose $a \in \mathbb{C}$ such that

$$e^a = c.$$

Then

$$f(z) = e^{G(z)+a}.$$

Therefore

$$g(z) = G(z) + a$$

is a holomorphic logarithm of f. □

Remark 6.24. The nonvanishing assumption is necessary. Since

$$e^w \neq 0$$

for every $w \in \mathbb{C}$, a function possessing a holomorphic logarithm can never vanish.

6.8 When Does a Holomorphic Logarithm Exist?

Simply connectedness is a convenient sufficient condition, but the exact criterion is expressed by contour integrals.

Theorem 6.25 (Criterion for a Holomorphic Logarithm)**.** *Let f be holomorphic and nonvanishing in a region Ω. Then f has a holomorphic logarithm in Ω if and only if*

$$\boxed{\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,\mathrm{d}z = 0}$$

for every closed curve $\gamma \subset \Omega$.

Proof. Suppose

$$f = e^g$$

for some holomorphic g. Then

$$\frac{f'}{f} = g'.$$

Hence for every closed curve γ,

$$\int_\gamma \frac{f'}{f}\,\mathrm{d}z = \int_\gamma g'(z)\,\mathrm{d}z = 0.$$

Conversely, suppose

$$\int_\gamma \frac{f'}{f}\,\mathrm{d}z = 0$$

for every closed curve. Then the integral of f'/f is path independent, so f'/f has a primitive G. The argument from the previous theorem then gives

$$f = e^{G+a}$$

for a suitable constant a. □

This criterion links the present chapter directly to the argument principle. For a closed curve γ,

$$\frac{1}{2\pi i}\int_\gamma \frac{f'}{f}\,\mathrm{d}z$$

measures the winding of $f(\gamma)$ around the origin.

Thus a global logarithm exists precisely when this winding obstruction vanishes for every closed curve in the domain.

6.9 Complex Powers

Once a branch of the logarithm has been chosen, complex powers can be defined.

Definition 6.26 (Complex Power)**.** Let L be a branch of the logarithm on a domain Ω. For $\alpha \in \mathbb{C}$, define

$$z^\alpha = e^{\alpha L(z)}.$$

For the principal branch,

$$z^\alpha = e^{\alpha z}, \qquad z \in \mathbb{C} \setminus (-\infty, 0].$$

Example 6.27 (Square Root)**.** Using the principal logarithm,

$$\sqrt{z} = e^{\frac{1}{2}z}.$$

If

$$z = re^{i\theta}, \qquad -\pi < \theta < \pi,$$

then

$$\sqrt{z} = \sqrt{r}\, e^{i\theta/2}.$$

Remark 6.28. Without choosing a branch,

$$z^\alpha$$

may be multivalued.

For example, solving

$$w^2 = z$$

for $z \neq 0$ gives two values differing by sign. A branch cut allows one of them to be selected continuously and holomorphically.

Remark 6.29. Familiar logarithmic identities require care in the complex plane.

For a fixed branch, it is not universally true that

$$(zw) = z + w.$$

The two sides may differ by an integer multiple of

$$2\pi i.$$

For example, using the principal branch,

$$(-i) = -\frac{\pi i}{2},$$

so

$$(-i) + (-i) = -\pi i.$$

But

$$(-i)^2 = -1,$$

and -1 lies on the branch cut of the standard principal logarithm. This illustrates why ordinary real logarithm identities cannot be used without checking the chosen domain and branch.

6.10 Topology Behind the Logarithm

The results of Chapters 2, 5, and 6 can now be combined into a single picture.

For

$$\Omega \subset \mathbb{C} \setminus \{0\},$$

the following conditions are closely related:

$$\boxed{\begin{array}{c} \text{a holomorphic branch of } \log z \text{ exists} \\ \Updownarrow \\ \dfrac{1}{z} \text{ has a primitive} \\ \Updownarrow \\ \displaystyle\int_\gamma \frac{\mathrm{d}z}{z} = 0 \quad \text{for every closed curve } \gamma \subset \Omega. \end{array}}$$

If Ω is simply connected, these conditions automatically hold.

On the other hand,

$$\mathbb{C} \setminus \{0\}$$

contains closed curves winding around the origin, and for such a curve

$$\frac{1}{2\pi i}\int_\gamma \frac{\mathrm{d}z}{z}$$

may be nonzero. This integer measures the winding of the curve around the origin.

Therefore

the obstruction to a global logarithm is topological.

Remark 6.30. This is an important conceptual transition in complex analysis.

Complex differentiability is a local property, but the existence of a single-valued logarithm is global. A logarithm exists locally near every nonzero point, yet it may fail to exist globally because the domain contains a hole.

The punctured plane

$$\mathbb{C}\setminus\{0\}$$

is the fundamental example:

$\frac{1}{z}$ is holomorphic locally, $\log z$ exists locally, but no global holomorphic logarithm exists.

6.11 Main Structure of the Chapter

The analytic side of the logarithm is

$z = \lvert z\rvert e^{i\theta}$ $\Downarrow$ θ is defined modulo 2π $\Downarrow$ $\log z = \log\lvert z\rvert + i\arg z$ is multivalued $\Downarrow$ choose a branch of the argument $\Downarrow$ obtain a single-valued holomorphic logarithm.

The topological side is

Ω simply connected $\Downarrow$ closed contour integrals of holomorphic functions vanish $\Downarrow$ $\frac{1}{z}$ has a primitive if $0 \notin \Omega$ $\Downarrow$ a holomorphic branch of $\log z$ exists.

More generally, if f is holomorphic and nonvanishing on a simply connected domain, then

$$\frac{f'}{f}$$

has a primitive, and therefore

$$\boxed{f = e^g}$$

for some holomorphic function g.

This is the precise analytic connection between topology, primitives, logarithms, and the logarithmic derivative.

Chapter 7

Entire Functions and Infinite Products

7 Entire Functions and Infinite Products

7.1 Entire Functions and Their Zeros

Definition 7.1 (Entire Function)**.** A function

$$f : \mathbb{C} \to \mathbb{C}$$

is called entire if it is holomorphic on the whole complex plane.

Basic examples include

$$e^z, \qquad \sin z, \qquad \cos z,$$

and every polynomial.

The global nature of an entire function makes its zeros particularly important. By the identity theorem, the zeros of a nonzero entire function cannot accumulate at a finite point.

Proposition 7.2. *Let f be an entire function which is not identically zero. Then every zero of f is isolated.*

Consequently, if

$$z_1, z_2, \dots$$

are distinct zeros of f, then

$$|z_n| \to \infty$$

whenever there are infinitely many zeros.

Remark 7.3. An entire function may have infinitely many zeros, but these zeros must escape to infinity.

For example,

$$\sin \pi z$$

has zeros at every integer:

$$0, \pm 1, \pm 2, \dots.$$

There is no finite accumulation point.

Example 7.4. The exponential function

$$e^z$$

has no zeros.

The function

$$\sin z$$

has zeros

$$z = n\pi, \qquad n \in \mathbb{Z},$$

all of which are simple.

7.2 Zeros and Growth

The zeros of an entire function and its growth at infinity are closely related.

A function with many zeros must, in a quantitative sense, grow sufficiently rapidly. Jensen's formula makes this relationship precise.

Theorem 7.5 (Jensen's Formula)**.** *Let f be holomorphic in a region containing the closed disc*

$$\{z : |z| \leq R\},$$

and suppose

$$f(0) \neq 0.$$

Let

$$z_1, \ldots, z_n$$

be the zeros of f in

$$|z| < R,$$

counted with multiplicity. Assume that f has no zeros on

$$|z| = R.$$

Then

$$\boxed{\log |f(0)| + \sum_{k=1}^{n} \log \frac{R}{|z_k|} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| \, \mathrm{d}\theta.}$$

Remark 7.6. Jensen's formula compares two quantities:

left-hand side : zeros inside the disc,
right-hand side : average size on the boundary.

Thus the distribution of zeros inside a disc is controlled by the growth of the function on its boundary.

Remark 7.7. A zero near the origin contributes more to the sum because

$$\log \frac{R}{|z_k|}$$

is larger when $|z_k|$ is smaller.

A zero close to the boundary contributes much less.

Jensen's formula therefore contains more information than simply counting zeros.

7.3 Counting Zeros

Definition 7.8 (Zero Counting Function)**.** For an entire function f, let

$$n_f(r)$$

denote the number of zeros of f in

$$|z| \leq r,$$

counted with multiplicity.

Jensen's formula implies that the growth of

$$n_f(r)$$

is constrained by the growth of f.

For example, if

$$M(r) = \max_{|z|=r} |f(z)|,$$

then Jensen's formula gives estimates relating

$$n_f(r)$$

to

$$\log M(r).$$

Remark 7.9. This gives a fundamental principle:

$$\boxed{\text{many zeros} \quad \Longrightarrow \quad \text{large growth.}}$$

Conversely, information about the growth of an entire function can restrict how densely its zeros may occur.

7.4 Order of Growth

To classify entire functions according to their growth, define their maximum modulus.

Definition 7.10 (Maximum Modulus)**.** For an entire function f, define

$$M(r, f) = \max_{|z|=r} |f(z)|.$$

When there is no ambiguity, we simply write

$$M(r).$$

Definition 7.11 (Finite Order)**.** An entire function f is said to be of finite order if there exist constants

$$A > 0, \qquad B > 0, \qquad \rho > 0$$

such that

$$|f(z)| \leq Ae^{B|z|^\rho}$$

for all sufficiently large $|z|$.

The order of an entire function is defined more precisely as follows.

Definition 7.12 (Order of an Entire Function)**.** For a nonconstant entire function f, its order is

$$\boxed{\rho = \limsup_{r\to\infty} \frac{\log\log M(r)}{\log r}.}$$

Example 7.13. For

$$f(z) = e^z,$$

we have

$$M(r) = e^r.$$

Therefore

$$\begin{aligned}\rho &= \limsup_{r\to\infty} \frac{\log r}{\log r} \\ &= 1.\end{aligned}$$

Thus e^z has order 1.

Example 7.14. For

$$f(z) = e^{z^m},$$

we have

$$M(r) = e^{r^m},$$

and hence

$$\rho = m.$$

Example 7.15. Every nonconstant polynomial has order 0.

Indeed, if

$$P(z) = a_n z^n + \cdots + a_0,$$

then

$$M(r)$$

grows polynomially rather than exponentially.

Remark 7.16. The order measures the exponential scale of growth:

$$\begin{aligned}\text{polynomial} &: \rho = 0, \\ e^z &: \rho = 1, \\ e^{z^2} &: \rho = 2, \\ e^{z^m} &: \rho = m.\end{aligned}$$

Thus the order is not the degree of an entire function. Rather, it measures how rapidly the maximum modulus grows as

$$|z| \to \infty.$$

7.5 From Polynomial Factorization to Infinite Products

The fundamental theorem of algebra gives the factorization

$$P(z) = c \prod_{k=1}^{n} (z - z_k)$$

for a polynomial with zeros

$$z_1, \ldots, z_n.$$

Equivalently, when none of the zeros is 0,

$$P(z) = C \prod_{k=1}^{n} \left(1 - \frac{z}{z_k}\right).$$

This suggests a natural question:

Can an entire function with infinitely many zeros be represented by an infinite product over its zeros?

A naive attempt would be

$$\prod_{n=1}^{\infty} \left(1 - \frac{z}{z_n}\right).$$

However, unlike a finite product, an infinite product need not converge.

7.6 Infinite Products

Definition 7.17 (Infinite Product)**.** Given complex numbers

$$a_1, a_2, \ldots,$$

the infinite product

$$\prod_{n=1}^{\infty}(1+a_n)$$

is defined as the limit of the partial products

$$P_N = \prod_{n=1}^{N}(1+a_n),$$

provided the limit exists.

For products of functions,

$$\prod_{n=1}^{\infty}(1+f_n(z)),$$

the relevant notion is local uniform convergence.

Theorem 7.18 (A Basic Convergence Criterion)**.** *Suppose that for every compact set $K \subset \Omega$,*

$$\sum_{n=1}^{\infty}|f_n(z)|$$

converges uniformly for $z \in K$. Then

$$\prod_{n=1}^{\infty}(1+f_n(z))$$

converges locally uniformly, provided the factors are eventually nonzero.

Remark 7.19. This is analogous to the familiar principle

$$\sum|a_n| < \infty \quad \Longrightarrow \quad \prod(1+a_n) \text{ converges.}$$

For products associated with zeros z_n, however,

$$\sum_n \left|\frac{z}{z_n}\right|$$

may diverge. Therefore the simple factors

$$1-\frac{z}{z_n}$$

are not always sufficient.

7.7 Why Naive Products Can Fail

Suppose

$$|z_n| \to \infty.$$

One might try

$$\prod_{n=1}^{\infty} \left(1 - \frac{z}{z_n}\right).$$

For fixed z,

$$\frac{z}{z_n} \to 0,$$

but this alone does not imply convergence of the product.

The problem is visible from the expansion

$$\log(1-w) = -w - \frac{w^2}{2} - \frac{w^3}{3} - \cdots.$$

If

$$w = \frac{z}{z_n},$$

the first-order term

$$-\frac{z}{z_n}$$

may produce a divergent series.

The idea of Weierstrass is to multiply each factor by a suitable exponential term which cancels the dangerous low-order terms.

7.8 Canonical Factors

Definition 7.20 (Weierstrass Canonical Factor)**.** For $p \in \mathbb{N}$, define

$$\boxed{E_p(w) = (1-w)\exp\left(w + \frac{w^2}{2} + \cdots + \frac{w^p}{p}\right).}$$

For $p = 0$, define

$$E_0(w) = 1 - w.$$

The first few canonical factors are

$$E_0(w) = 1 - w,$$

$$E_1(w) = (1-w)e^{w},$$

and

$$E_2(w) = (1-w)e^{w+w^2/2}.$$

Proposition 7.21. *Near $w = 0$,*

$$\log E_p(w) = -\frac{w^{p+1}}{p+1} - \frac{w^{p+2}}{p+2} - \cdots.$$

In particular,

$$\log E_p(w) = O(w^{p+1}).$$

Proof. For $|w| < 1$,

$$\log(1-w) = -w - \frac{w^2}{2} - \cdots .$$

Therefore

$$\begin{aligned}\log E_p(w) &= \log(1-w) + w + \frac{w^2}{2} + \cdots + \frac{w^p}{p} \\ &= -\frac{w^{p+1}}{p+1} - \frac{w^{p+2}}{p+2} - \cdots .\end{aligned}$$

□

Remark 7.22. This is the essential reason canonical factors work.

The factor

$$1 - w$$

creates the desired zero at

$$w = 1,$$

while

$$\exp\left(w + \frac{w^2}{2} + \cdots + \frac{w^p}{p}\right)$$

does not create any new zeros.

Its only role is to improve convergence by cancelling the first p terms of

$$\log(1-w).$$

Thus:

canonical factor = zero-producing factor × convergence correction.

7.9 Weierstrass Product Theorem

The canonical factors allow one to construct entire functions with prescribed zeros.

Theorem 7.23 (Weierstrass Product Theorem)**.** *Let*

$$z_1, z_2, \ldots$$

be a sequence of nonzero complex numbers such that

$$|z_n| \to \infty.$$

Let

$$m_1, m_2, \ldots$$

be prescribed positive integer multiplicities.

Then there exist nonnegative integers

$$p_1, p_2, \ldots$$

such that

$$\boxed{P(z) = \prod_{n=1}^{\infty} E_{p_n}\left(\frac{z}{z_n}\right)^{m_n}}$$

converges locally uniformly and defines an entire function whose zeros are exactly the points z_n, with multiplicities m_n.

If a zero of multiplicity m is also required at the origin, one simply multiplies by

$$z^m.$$

Thus a general Weierstrass product has the form

$$z^m \prod_{n=1}^{\infty} E_{p_n}\left(\frac{z}{z_n}\right)^{m_n}.$$

Remark 7.24. This theorem is the infinite analogue of polynomial factorization.

For a polynomial,

$$P(z) = C\prod_{k=1}^{n}(z - z_k).$$

For an entire function, infinitely many zeros may occur, so convergence corrections must be inserted:

finite polynomial product $\longrightarrow$ Weierstrass canonical product.

7.10 Factorization of an Entire Function

The Weierstrass product theorem constructs an entire function from a given zero set. We can also apply it to the zeros of an already existing entire function.

Theorem 7.25 (Weierstrass Factorization)**.** *Let f be an entire function, not identically zero. Suppose f has a zero of order m at 0, and let*

$$z_1, z_2, \ldots$$

be its nonzero zeros, counted with multiplicity.

Then

$$\boxed{f(z) = z^m e^{g(z)} \prod_{n=1}^{\infty} E_{p_n}\left(\frac{z}{z_n}\right),}$$

for some entire function g and suitable nonnegative integers p_n.

Remark 7.26. Why does the extra factor

$$e^{g(z)}$$

appear?

Suppose $P(z)$ is a Weierstrass product with exactly the same zeros as f, with the same multiplicities. Then

$$\frac{f(z)}{P(z)}$$

is entire and never zero.

Since $\mathbb{C}$ is simply connected, Chapter 6 implies that every nonvanishing entire function has a holomorphic logarithm. Hence

$$\frac{f(z)}{P(z)} = e^{g(z)}$$

for some entire g.

Therefore

$$f(z) = e^{g(z)}P(z).$$

This is exactly where the theory of holomorphic logarithms from Chapter 6 enters the factorization theory of entire functions.

7.11 Example: The Sine Function

The zeros of

$$\sin \pi z$$

are exactly the integers

$$0, \pm 1, \pm 2, \dots.$$

Its product representation is

$$\boxed{\frac{\sin \pi z}{\pi z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).}$$

Equivalently,

$$\boxed{\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).}$$

Remark 7.27. This formula is one of the clearest examples of the philosophy of Weierstrass factorization.

Compare

$$P(z) = C \prod_{k=1}^{N} (z - z_k)$$

for a polynomial with

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

The infinitely many zeros

$$0, \pm 1, \pm 2, \dots$$

are encoded directly in the infinite product.

Remark 7.28. Pairing the zeros n and $-n$ is important:

$$\left(1 - \frac{z}{n}\right)\left(1 + \frac{z}{n}\right) = 1 - \frac{z^2}{n^2}.$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty,$$

the paired product has much better convergence than treating the two sequences separately with naive linear factors.

7.12 Example: Recovering an Infinite Series

The product formula for sine also illustrates how factorization can lead to identities involving infinite series.

Starting from

$$\frac{\sin \pi z}{\pi z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right),$$

take the logarithmic derivative:

$$\frac{\mathrm{d}}{\mathrm{d}z} \log\left(\frac{\sin \pi z}{\pi z}\right) = \sum_{n=1}^{\infty} \frac{\mathrm{d}}{\mathrm{d}z} \log\left(1 - \frac{z^2}{n^2}\right).$$

Hence

$$\pi \cot(\pi z) - \frac{1}{z} = \sum_{n=1}^{\infty} \frac{-2z}{n^2 - z^2}.$$

Equivalently,

$$\boxed{\pi \cot(\pi z) = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}.}$$

Remark 7.29. This identity illustrates a recurring theme:

$$\boxed{\text{zeros} \to \text{infinite product} \to \text{logarithmic derivative} \to \text{infinite series.}}$$

The logarithmic derivative

$$\frac{f'}{f}$$

appears again because it converts multiplicative information into additive information.

7.13 Finite Order and Canonical Products

For arbitrary entire functions, the integers p_n in a Weierstrass product may vary with n.

For entire functions of finite order, much more can be said.

Definition 7.30 (Exponent of Convergence of Zeros)**.** Let

$$z_1, z_2, \ldots$$

be the nonzero zeros of an entire function, repeated according to multiplicity.

The exponent of convergence of the zeros is

$$\lambda = \inf \left\{ s > 0 : \sum_{n=1}^{\infty} \frac{1}{|z_n|^s} < \infty \right\}.$$

Remark 7.31. The exponent of convergence measures how densely the zeros escape to infinity.

If the zeros grow rapidly, then

$$\sum \frac{1}{|z_n|^s}$$

converges for relatively small s.

If the zeros are dense, a larger exponent is required.

For an entire function of finite order ρ, the zero set cannot be arbitrarily dense. In particular, its exponent of convergence satisfies

$$\lambda \leq \rho.$$

This is another precise form of the principle

$$\text{density of zeros} \leq \text{growth of the function.}$$

7.14 Genus

Definition 7.32 (Genus of a Canonical Product)**.** Suppose

$$P(z) = \prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

converges for some fixed nonnegative integer p.

The smallest such p, when it exists, is related to the genus of the canonical product.

For finite-order entire functions, only finitely many correction terms are needed. This is what allows the general Weierstrass factorization theorem to be sharpened into Hadamard's theorem.

7.15 Hadamard Factorization Theorem

Theorem 7.33 (Hadamard Factorization Theorem)**.** *Let f be an entire function of finite order*

$$\rho < \infty.$$

Suppose f has a zero of multiplicity m at 0, and let

$$z_1, z_2, \ldots$$

be its nonzero zeros, repeated according to multiplicity.

Then

$$\boxed{f(z) = z^m e^{Q(z)} \prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right),}$$

where Q is a polynomial and p may be chosen as a finite nonnegative integer compatible with the convergence of the zero set. In particular, the factorization can be chosen so that its genus does not exceed the order of f.

Remark 7.34. Compare the general Weierstrass factorization

$$f(z) = z^m e^{g(z)} \prod E_{p_n}\left(\frac{z}{z_n}\right)$$

with Hadamard's finite-order factorization

$$f(z) = z^m e^{Q(z)} \prod E_p\left(\frac{z}{z_n}\right).$$

The important improvement is

$$\boxed{\begin{array}{l} g(z) \longrightarrow Q(z) \text{ is a polynomial,} \\ \quad p_n \longrightarrow \text{a finite genus can be used.} \end{array}}$$

Thus a growth condition imposes strong algebraic structure on the factorization.

7.16 Examples of Hadamard Factorization

Example 7.35. Consider

$$f(z) = e^z.$$

It has no zeros and has order 1.

Its Hadamard factorization therefore consists only of the exponential factor:

$$e^z = e^{Q(z)},$$

where

$$Q(z) = z.$$

Example 7.36. The function

$$\sin \pi z$$

has order 1 and zeros at all integers.

Its factorization

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

shows explicitly how a finite-order entire function is determined, up to an exponential polynomial factor, by its zeros.

7.17 What the Zeros Determine

Suppose f and g are entire functions having exactly the same zeros, with the same multiplicities.

Then

$$\frac{f}{g}$$

is entire and never zero. By the logarithm theorem from Chapter 6,

$$\frac{f(z)}{g(z)} = e^{h(z)}$$

for some entire function h.

Therefore

$$\boxed{f(z) = e^{h(z)} g(z).}$$

Remark 7.37. The zeros of an entire function do not determine the function uniquely.

For example,

$$\sin z$$

and

$$e^z \sin z$$

have exactly the same zeros, with the same multiplicities.

The missing information is a nonvanishing entire factor

$$e^{h(z)}.$$

For finite-order entire functions, Hadamard's theorem restricts this factor to

$$e^{Q(z)}$$

with Q a polynomial.

7.18 Three Levels of Factorization

It is useful to compare three increasingly general situations.

1. **Polynomial factorization.**
   If P is a polynomial,
   $$P(z) = C \prod_{k=1}^{n} (z - z_k).$$

2. **Weierstrass factorization.**
   If f is an arbitrary nonzero entire function,
   $$f(z) = z^m e^{g(z)} \prod_{n=1}^{\infty} E_{p_n}\left(\frac{z}{z_n}\right).$$

3. **Hadamard factorization.**
   If f is entire of finite order,
   $$f(z) = z^m e^{Q(z)} \prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right),$$
   where the exponential correction is polynomial and the canonical product has finite genus.

Thus

$$\boxed{\text{polynomials} \subset \text{finite-order entire functions} \subset \text{entire functions},}$$

with a corresponding hierarchy of factorization theorems.

7.19 Main Structure of the Chapter

The first main theme of this chapter is the relation between zeros and growth:

$$\boxed{\begin{array}{c}\text{entire function}\\ \Downarrow\\ \text{isolated zeros}\\ \Downarrow\\ \text{Jensen's formula}\\ \Downarrow\\ \text{zero distribution is controlled by growth}\\ \Downarrow\\ \text{order of an entire function.}\end{array}}$$

The second main theme is factorization:

$$\boxed{\begin{array}{c}\text{prescribed zeros}\\ \Downarrow\\ \text{naive infinite product may diverge}\\ \Downarrow\\ \text{canonical factors } E_p\\ \Downarrow\\ \text{Weierstrass product}\\ \Downarrow\\ \text{factorization of entire functions.}\end{array}}$$

Finally, imposing finite order gives the stronger conclusion

$$\boxed{\text{finite growth} + \text{zeros} \quad \Longrightarrow \quad \text{Hadamard factorization.}}$$

Conceptually, the progression from polynomial factorization to entire functions is

$$\boxed{\begin{array}{rcl}\text{Polynomial} & : & C\prod_{k=1}^{n}(z-z_k),\\ \text{Entire} & : & e^{g(z)}\prod_{n=1}^{\infty}E_{p_n}\left(\frac{z}{z_n}\right),\\ \text{Finite order} & : & e^{Q(z)}\prod_{n=1}^{\infty}E_{p}\left(\frac{z}{z_n}\right).\end{array}}$$

The zeros provide the multiplicative skeleton of an entire function, while its growth determines how complicated the remaining zero-free factor is.

Chapter 8

Conformal Mappings

8 Conformal Mappings

8.1 Conformal Equivalence

Complex analysis is not only an analytic theory but also a geometric one. Holomorphic functions with nonzero derivative preserve angles locally, and this leads to the theory of conformal mappings.

Definition 8.1 (Conformal Map)**.** Let $U, V \subset \mathbb{C}$ be domains. A map

$$f : U \to V$$

is called conformal if it is bijective and holomorphic.

Two domains U and V are called conformally equivalent if there exists a conformal map

$$f : U \to V.$$

Such a map is also called a biholomorphism.

Remark 8.2. In many texts, "conformal" is used locally to mean angle-preserving with nonzero derivative. In these notes, when discussing equivalence of domains, we use the stronger global meaning:

bijective and holomorphic.

Theorem 8.3 (Holomorphic Inverse)**.** *Suppose*

$$f : U \to V$$

is bijective and holomorphic. Then

$$f'(z) \neq 0$$

for every $z \in U$, and the inverse

$$f^{-1} : V \to U$$

is holomorphic.

Moreover,

$$\boxed{(f^{-1})'(w) = \frac{1}{f'(f^{-1}(w))}.}$$

Proof. Suppose

$$f'(z_0) = 0.$$

Since f is nonconstant and holomorphic, its local power series at z_0 has the form

$$f(z) - f(z_0) = a_m(z - z_0)^m + \cdots$$

for some

$$m \geq 2.$$

Locally this behaves like

$$(z - z_0)^m,$$

which is not one-to-one near z_0. This contradicts injectivity. Therefore

$$f'(z) \neq 0.$$

The inverse function theorem then gives that f^{-1} is holomorphic, and differentiating

$$f(f^{-1}(w)) = w$$

gives

$$f'(f^{-1}(w))(f^{-1})'(w) = 1.$$

□

Corollary 8.4. *Two domains U and V are conformally equivalent if and only if there exist holomorphic maps*

$$f : U \to V, \qquad g : V \to U$$

such that

$$g(f(z)) = z$$

for every $z \in U$, and

$$f(g(w)) = w$$

for every $w \in V$.

8.2 Why Holomorphic Maps Preserve Angles

Suppose f is holomorphic at z_0 and

$$f'(z_0) \neq 0.$$

Then locally,

$$f(z_0 + h) = f(z_0) + f'(z_0)h + o(|h|).$$

Write

$$f'(z_0) = re^{i\theta}, \qquad r > 0.$$

Multiplication by

$$re^{i\theta}$$

has two geometric effects:

$$h \longmapsto re^{i\theta}h.$$

It rotates by the angle θ and scales lengths by r.

Therefore infinitesimally,

$$\boxed{\text{holomorphic map with } f'(z_0) \neq 0 = \text{rotation} + \text{dilation.}}$$

Remark 8.5. Rotations and dilations preserve angles. This explains geometrically why holomorphic functions with nonzero derivative are conformal.

The condition

$$f'(z_0) \neq 0$$

is essential. For example,

$$f(z) = z^2$$

has

$$f'(0) = 0,$$

and near 0 it doubles angles rather than preserving them.

8.3 Standard Domains

Two domains appear repeatedly in conformal mapping problems.

Definition 8.6 (Unit Disc)**.** The unit disc is

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

Definition 8.7 (Upper Half-Plane)**.** The upper half-plane is

$$\mathbb{H} = \{z \in \mathbb{C} : \Im z > 0\}.$$

These are the two principal model domains of one-variable complex analysis.

8.4 Elementary Conformal Mappings

Many conformal mapping problems can be solved by composing a small collection of elementary maps.

8.4.1 Translations

A translation has the form

$$f(z) = z + a, \qquad a \in \mathbb{C}.$$

Its inverse is

$$f^{-1}(w) = w - a.$$

Translations move domains without changing their shape or orientation.

8.4.2 Rotations and Dilations

For

$$c \in \mathbb{C} \setminus \{0\},$$

consider

$$f(z) = cz.$$

If

$$c = re^{i\theta},$$

then f dilates by r and rotates by θ.

In particular, if

$$|c| = 1,$$

then

$$f(z) = e^{i\theta} z$$

is a pure rotation.

8.4.3 Power Maps

The map

$$f(z) = z^{\alpha}$$

can be used to change the angle of a sector.

For example, let

$$S_{\alpha} = \{z : 0 < \arg z < \alpha\}.$$

Then

$$f(z) = z^{\pi/\alpha}$$

maps S_{α} conformally onto the upper half-plane.

Example 8.8. The map

$$f(z) = z^2$$

takes the first quadrant

$$0 < \arg z < \frac{\pi}{2}$$

onto

$$\mathbb{H}.$$

Indeed,

$$0 < \arg(z^2) < \pi.$$

Remark 8.9. When the exponent is not an integer, a branch of the logarithm must first be chosen:

$$z^\alpha = e^{\alpha z}.$$

Thus the branch theory of Chapter 6 is directly used in conformal mapping.

8.4.4 The Exponential Map

Write

$$z = x + iy.$$

Then

$$e^z = e^x e^{iy}.$$

Thus

$$|e^z| = e^x, \qquad \arg(e^z) = y.$$

Consequently, horizontal strips are mapped to sectors.

Example 8.10. The strip

$$S = \{z = x + iy : 0 < y < \pi\}$$

is mapped conformally by

$$f(z) = e^z$$

onto the upper half-plane.

Indeed,

$$0 < y < \pi$$

implies

$$0 < \arg(e^z) < \pi.$$

Similarly,

$$-\frac{\pi}{2} < \Im z < \frac{\pi}{2}$$

is mapped by e^z onto the right half-plane.

8.4.5 The Logarithm

The logarithm performs the inverse operation.

If

$$z = re^{i\theta},$$

then

$$z = \log r + i\theta.$$

Thus angle becomes imaginary part.

For example, the upper half-plane

$$0 < \arg z < \pi$$

is mapped by a suitable branch of $\log z$ onto the horizontal strip

$$\{w = u + iv : 0 < v < \pi\}.$$

Remark 8.11. This gives the useful correspondence

$$\boxed{\text{sector} \xrightarrow{\log} \text{strip}, \qquad \text{strip} \xrightarrow{\exp} \text{sector}.}$$

8.5 Fractional Linear Transformations

Definition 8.12 (Fractional Linear Transformation)**.** A fractional linear transformation, or Möbius transformation, is a map

$$f(z) = \frac{az + b}{cz + d},$$

where

$$a, b, c, d \in \mathbb{C}$$

and

$$ad - bc \neq 0.$$

The condition

$$ad - bc \neq 0$$

ensures that the map is nonconstant and invertible on the extended complex plane.

Proposition 8.13. *A Möbius transformation is conformal wherever it is finite.*

Indeed,

$$f'(z) = \frac{ad - bc}{(cz + d)^2},$$

which is never zero.

Remark 8.14. Möbius transformations map generalized circles to generalized circles, where a generalized circle means either a circle or a straight line.

This makes them particularly useful for mapping discs and half-planes.

8.6 The Unit Disc and the Upper Half-Plane

The upper half-plane and the unit disc are conformally equivalent.

Theorem 8.15. *The map*

$$F(z) = \frac{z-i}{z+i}$$

maps

$$\mathbb{H}$$

conformally onto

$$\mathbb{D}.$$

Its inverse is

$$F^{-1}(w) = i\frac{1+w}{1-w}.$$

Proof. Let

$$z = x + iy, \qquad y > 0.$$

Then

$$\begin{aligned} |z+i|^2 - |z-i|^2 &= x^2 + (y+1)^2 - \left[x^2 + (y-1)^2\right] \\ &= 4y > 0. \end{aligned}$$

Hence

$$|z-i| < |z+i|,$$

so

$$\left|\frac{z-i}{z+i}\right| < 1.$$

Thus

$$F(\mathbb{H}) \subset \mathbb{D}.$$

The explicit inverse proves bijectivity. □

Remark 8.16. This transformation is worth remembering because it allows results about $\mathbb{D}$ to be transferred to $\mathbb{H}$, and conversely.

8.7 How to Construct a Conformal Mapping

A practical conformal-mapping problem is often solved by a sequence of simple steps.

For example, to map a complicated sector or strip onto $\mathbb{D}$, one may use

$$\boxed{\text{original domain} \longrightarrow \text{sector} \longrightarrow \mathbb{H} \longrightarrow \mathbb{D}.}$$

Typical tools are:

1. translation:
$$z \mapsto z - a;$$
2. rotation and dilation:
$$z \mapsto cz;$$
3. power maps:
$$z \mapsto z^{\alpha};$$
4. logarithm:
$$z \mapsto z;$$

5. exponential:
$$z \mapsto e^z;$$
6. Möbius transformations:
$$z \mapsto \frac{az+b}{cz+d}.$$

Remark 8.17. It is often much easier to construct
$$f = f_n \circ \cdots \circ f_2 \circ f_1$$
than to guess one complicated formula directly.

At each stage, draw or identify the image of the boundary. This usually determines the image domain.

8.8 Harmonic Functions and Conformal Maps

Definition 8.18 (Harmonic Function)**.** A twice continuously differentiable real-valued function
$$u = u(x, y)$$
is harmonic if
$$\boxed{\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.}$$

If
$$f = u + iv$$
is holomorphic, then both u and v are harmonic.

Proposition 8.19. *Let*
$$F : V \to U$$
be holomorphic, and let
$$u : U \to \mathbb{R}$$
be harmonic. Then
$$u \circ F$$
is harmonic on V*.*

Remark 8.20. Harmonicity is preserved under conformal change of variables.

This is one reason conformal mappings are useful in potential theory and boundary-value problems: one may transform a complicated domain into a simpler one, solve the harmonic problem there, and transform the solution back.

8.9 Schwarz's Lemma

The unit disc has a particularly rigid holomorphic geometry.

Theorem 8.21 (Schwarz's Lemma)**.** *Let*
$$f : \mathbb{D} \to \mathbb{D}$$
be holomorphic and suppose
$$f(0) = 0.$$
Then:

1.

$$|f(z)| \leq |z| \qquad \textit{for every } z \in \mathbb{D};$$

2.

$$|f'(0)| \leq 1;$$

3. if there exists $z_0 \neq 0$ *such that*

$$|f(z_0)| = |z_0|,$$

or if

$$|f'(0)| = 1,$$

then

$$f(z) = e^{i\theta} z$$

for some

$$\theta \in \mathbb{R}.$$

Proof. Since

$$f(0) = 0,$$

write

$$f(z) = zg(z),$$

where g is holomorphic on $\mathbb{D}$.

Fix

$$0 < r < 1.$$

On

$$|z| = r,$$

we have

$$|g(z)| = \frac{|f(z)|}{r} < \frac{1}{r}.$$

By the maximum modulus principle,

$$|g(z)| \leq \frac{1}{r}$$

for

$$|z| \leq r.$$

Letting

$$r \to 1$$

gives

$$|g(z)| \leq 1.$$

Therefore

$$|f(z)| \leq |z|.$$

Since

$$f'(0) = g(0),$$

we obtain

$$|f'(0)| \leq 1.$$

If equality occurs at a nonzero point or at the derivative, then g attains its maximum modulus 1 inside the disc. By the maximum modulus principle, g is constant:

$$g(z) = e^{i\theta}.$$

Hence

$$f(z) = e^{i\theta} z.$$

□

Remark 8.22. Schwarz's lemma says that a holomorphic self-map of the unit disc fixing the origin cannot expand distances from the origin.

The only maps that preserve the size exactly are rotations.

8.10 Automorphisms of the Unit Disc

Definition 8.23 (Automorphism)**.** Let $\Omega \subset \mathbb{C}$ be a domain. An automorphism of Ω is a conformal map

$$f : \Omega \to \Omega.$$

The set of all automorphisms is denoted by

$$\mathrm{Aut}(\Omega).$$

If $f, g \in \mathrm{Aut}(\Omega)$, then

$$f \circ g \in \mathrm{Aut}(\Omega),$$

and

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}.$$

Thus

$$\mathrm{Aut}(\Omega)$$

forms a group under composition.

For

$$\alpha \in \mathbb{D},$$

define

$$\varphi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}.$$

Proposition 8.24. *The map*

$$\varphi_\alpha$$

is an automorphism of $\mathbb{D}$*, and*

$$\varphi_\alpha(\alpha) = 0, \qquad \varphi_\alpha(0) = \alpha.$$

Moreover,

$$\varphi_\alpha \circ \varphi_\alpha = \mathrm{id}\,.$$

Theorem 8.25 (Automorphisms of the Disc)**.** *Every automorphism of* $\mathbb{D}$ *has the form*

$$\boxed{f(z) = e^{i\theta} \frac{\alpha - z}{1 - \overline{\alpha} z},}$$

where

$$\theta \in \mathbb{R}, \qquad \alpha \in \mathbb{D}.$$

Proof. Let

$$\alpha = f^{-1}(0).$$

Then

$$f(\alpha) = 0.$$

Since

$$\varphi_\alpha(0) = \alpha,$$

the map

$$g = f \circ \varphi_\alpha$$

is an automorphism of $\mathbb{D}$ satisfying

$$g(0) = 0.$$

By Schwarz's lemma,

$$|g(z)| \leq |z|.$$

Applying Schwarz's lemma to g^{-1} gives the reverse inequality, so

$$|g(z)| = |z|.$$

Hence equality holds in Schwarz's lemma and

$$g(z) = e^{i\theta} z.$$

Therefore

$$f(z) = e^{i\theta} \varphi_\alpha(z),$$

up to the equivalent choice of sign absorbed into the rotation. □

Corollary 8.26. *The only automorphisms of* $\mathbb{D}$ *fixing* 0 *are rotations:*

$$f(z) = e^{i\theta} z.$$

8.11 Transitivity of the Disc Automorphism Group

Definition 8.27 (Transitivity)**.** A group of automorphisms acts transitively on a domain if, for any

$$\alpha, \beta \in \Omega,$$

there exists an automorphism ψ such that

$$\psi(\alpha) = \beta.$$

Proposition 8.28. *The automorphism group*

$$\mathrm{Aut}(\mathbb{D})$$

acts transitively on $\mathbb{D}$*.*

Indeed, one may use the composition

$$\psi = \varphi_\beta \circ \varphi_\alpha$$

with an appropriate rotation if necessary.

Remark 8.29. No point inside the unit disc is geometrically distinguished from another from the viewpoint of conformal geometry.

Any point can be moved to any other point by an automorphism.

8.12 Automorphisms of the Upper Half-Plane

The automorphisms of $\mathbb{H}$ have an equally explicit description.

Definition 8.30 (Special Linear Group)**.** Define

$$SL_2(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R},\ ad - bc = 1 \right\}.$$

For

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R}),$$

define

$$f_M(z) = \frac{az + b}{cz + d}.$$

Theorem 8.31 (Automorphisms of the Upper Half-Plane)**.** *Every automorphism of*

$$\mathbb{H}$$

has the form

$$\boxed{f(z) = \frac{az + b}{cz + d},}$$

where

$$a, b, c, d \in \mathbb{R}$$

and

$$ad - bc > 0.$$

After multiplying all coefficients by a common nonzero constant, one may normalize

$$ad - bc = 1.$$

Proposition 8.32. *If*

$$M \in SL_2(\mathbb{R}),$$

then

$$f_M^{-1} = f_{M^{-1}},$$

and

$$f_M \circ f_N = f_{MN}.$$

Definition 8.33 (Projective Special Linear Group)**.** Since

$$f_M = f_{-M},$$

the automorphism group of the upper half-plane is naturally identified with the projective special linear group

$$PSL_2(\mathbb{R}) = SL_2(\mathbb{R})/\{I, -I\}.$$

Remark 8.34. This is a useful point where complex analysis meets group theory:

$$\boxed{\mathrm{Aut}(\mathbb{H}) \cong PSL_2(\mathbb{R}).}$$

8.13 Conjugating Automorphism Groups

Let

$$F : \mathbb{H} \to \mathbb{D}$$

be a conformal equivalence.

If

$$\varphi \in \mathrm{Aut}(\mathbb{D}),$$

then

$$F^{-1} \circ \varphi \circ F$$

is an automorphism of $\mathbb{H}$.

Conversely, if

$$\psi \in \mathrm{Aut}(\mathbb{H}),$$

then

$$F \circ \psi \circ F^{-1}$$

is an automorphism of $\mathbb{D}$.

Thus the two automorphism groups are related by conjugation.

Remark 8.35. This is a general principle:

if

$$U \quad \text{and} \quad V$$

are conformally equivalent, then their automorphism groups carry the same structure after conjugation by a conformal equivalence.

8.14 The Riemann Mapping Theorem

The preceding examples show that many apparently different planar domains are conformally equivalent. The Riemann mapping theorem gives an extremely general version of this phenomenon.

Definition 8.36 (Proper Domain)**.** A domain

$$\Omega \subset \mathbb{C}$$

is called proper if

$$\Omega \neq \varnothing$$

and

$$\Omega \neq \mathbb{C}.$$

Theorem 8.37 (Riemann Mapping Theorem)**.** *Let*

$$\Omega \subsetneq \mathbb{C}$$

be a nonempty simply connected domain.

For any

$$z_0 \in \Omega,$$

there exists a unique conformal map

$$F : \Omega \to \mathbb{D}$$

such that

$$F(z_0) = 0$$

and

$$F'(z_0) > 0.$$

Remark 8.38. The normalization

$$F(z_0) = 0, \qquad F'(z_0) > 0$$

is required for uniqueness.

Without this normalization, one may compose F with any automorphism of the unit disc and obtain another conformal map.

Corollary 8.39. *Any two proper simply connected domains in* $\mathbb{C}$ *are conformally equivalent.*

Proof. Let

$$\Omega_1, \Omega_2$$

be proper simply connected domains. By the Riemann mapping theorem, there exist conformal maps

$$F_1 : \Omega_1 \to \mathbb{D}$$

and

$$F_2 : \Omega_2 \to \mathbb{D}.$$

Therefore

$$F_2^{-1} \circ F_1 : \Omega_1 \to \Omega_2$$

is a conformal equivalence. □

Remark 8.40. This theorem is remarkable because the boundary of Ω may be very complicated. As long as the domain is simply connected and is not the whole complex plane, its interior conformal geometry is equivalent to that of the unit disc.

Thus from the conformal viewpoint, an enormous class of planar domains are essentially the same.

8.15 Why the Whole Plane Is Excluded

The condition

$$\Omega \neq \mathbb{C}$$

cannot be removed.

Suppose there were a conformal map

$$f : \mathbb{C} \to \mathbb{D}.$$

Then f would be entire and bounded:

$$|f(z)| < 1.$$

By Liouville's theorem,

$$f$$

would be constant, contradicting bijectivity.

Therefore

$$\mathbb{C}$$

is not conformally equivalent to $\mathbb{D}$.

Remark 8.41. This is a good example of how earlier analytic rigidity results reappear in conformal geometry:

$$\boxed{\text{Liouville's theorem} \Longrightarrow \mathbb{C} \ncong \mathbb{D}.}$$

8.16 Normal Families

The proof of the Riemann mapping theorem relies on compactness properties of families of holomorphic functions.

Definition 8.42 (Locally Uniformly Bounded)**.** A family of holomorphic functions

$$\mathcal{F}$$

on Ω is locally uniformly bounded if for every compact set

$$K \subset \Omega,$$

there exists

$$M_K > 0$$

such that

$$|f(z)| \leq M_K$$

for every

$$z \in K$$

and every

$$f \in \mathcal{F}.$$

Definition 8.43 (Equicontinuity)**.** A family $\mathcal{F}$ is equicontinuous on a compact set K if for every

$$\varepsilon > 0$$

there exists

$$\delta > 0$$

such that

$$|z - w| < \delta$$

implies

$$|f(z) - f(w)| < \varepsilon$$

for every

$$f \in \mathcal{F}.$$

Proposition 8.44. *A locally uniformly bounded family of holomorphic functions is equicontinuous on every compact subset of its domain.*

Remark 8.45. The reason is Cauchy's inequality: local bounds on the functions produce local bounds on their derivatives.

Then

$$|f(z) - f(w)|$$

can be controlled uniformly using the derivative bound.

Definition 8.46 (Normal Family)**.** A family $\mathcal{F}$ of holomorphic functions on Ω is normal if every sequence

$$\{f_n\} \subset \mathcal{F}$$

has a subsequence converging uniformly on every compact subset of Ω.

Theorem 8.47 (Montel's Theorem)**.** *A locally uniformly bounded family of holomorphic functions on a domain Ω is normal.*

Remark 8.48. The role of normal families is compactness.

Instead of manipulating an infinite sequence of candidate conformal maps directly, one extracts a subsequence

$$f_{n_k}$$

which converges locally uniformly to a holomorphic function.

This compactness argument is one of the main ingredients in the proof of the Riemann mapping theorem.

8.17 Limits of Injective Holomorphic Functions

Proposition 8.49. *Let Ω be connected and open. Suppose*

$$f_n : \Omega \to \mathbb{C}$$

are holomorphic and injective, and

$$f_n \to f$$

uniformly on every compact subset of Ω.

Then f is either injective or constant.

Remark 8.50. Thus injectivity is almost preserved under locally uniform limits.

The only possible degeneration is that the entire family collapses to a constant map.

8.18 Idea Behind the Riemann Mapping Theorem

The complete proof of the Riemann mapping theorem is technically longer than the core undergraduate results developed earlier, but its main strategy can be summarized as follows.

Fix

$$z_0 \in \Omega.$$

Consider the family of injective holomorphic maps

$$f : \Omega \to \mathbb{D}$$

satisfying

$$f(z_0) = 0.$$

Among these maps, seek one maximizing

$$|f'(z_0)|.$$

Using normal-family compactness, one extracts a locally uniformly convergent subsequence of maps whose derivatives approach the supremum.

The limit is holomorphic and remains injective unless it degenerates to a constant. The normalization prevents this degeneration.

One then proves that the extremal map must be onto $\mathbb{D}$; otherwise one could modify it to obtain another admissible map with larger derivative at z_0, contradicting extremality.

Remark 8.51. The proof therefore has an extremal structure:

construct a family of candidate maps
$\Downarrow$
maximize $\lvert f'(z_0)\rvert$
$\Downarrow$
use normal-family compactness
$\Downarrow$
obtain an extremal map
$\Downarrow$
prove the extremal map is onto.

This is more important conceptually than memorizing every technical step of the proof.

8.19 Conformal Mappings onto Polygonal Regions

Definition 8.52 (Polygonal Region)**.** A polygonal region P is a bounded simply connected open set whose boundary is a polygonal line.

By the Riemann mapping theorem, every polygonal region is conformally equivalent to the unit disc.

Under suitable boundary regularity, a conformal map

$$F : \mathbb{D} \to P$$

extends continuously to the boundary, giving a correspondence between

$$\partial\mathbb{D}$$

and

$$\partial P.$$

Remark 8.53. The existence of a conformal map onto a polygon is guaranteed abstractly by the Riemann mapping theorem.

Finding an explicit formula is a separate and substantially more difficult problem. In applications, such maps are closely related to Schwarz–Christoffel transformations.

8.20 A Practical Mapping Dictionary

The most useful transformations in elementary problems can be summarized as follows:

$z \mapsto z + a$	:	translation,
$z \mapsto cz$	:	rotation and dilation,
$z \mapsto z^{\alpha}$	:	sector angle change,
$z \mapsto z$	:	sector $\to$ strip,
$z \mapsto e^{z}$	:	strip $\to$ sector,
$z \mapsto \dfrac{az+b}{cz+d}$	:	circles/lines, discs/half-planes.

A typical mapping problem can therefore be approached by asking:

1. Can the domain first be translated or rotated into a standard position?
2. Is it a sector? If so, use a power map.
3. Is it a strip? If so, use the exponential.
4. Would a logarithm turn the region into a strip?
5. Once a half-plane is obtained, use a Möbius transformation to reach the unit disc.

8.21 Main Structure of the Chapter

The elementary geometric part of the theory is

$$\boxed{\begin{array}{c}\text{holomorphic} + f' \neq 0 \\ \Downarrow \\ \text{local rotation and dilation} \\ \Downarrow \\ \text{angle preservation} \\ \Downarrow \\ \text{conformal mapping.}\end{array}}$$

The standard mapping tools are

$$\boxed{\text{translation} \to \text{rotation/dilation} \to \text{power/log/exp} \to \text{Möbius transformation.}}$$

The geometry of the unit disc is controlled by

$$\boxed{\text{Schwarz lemma} \Longrightarrow \mathrm{Aut}(\mathbb{D}) = \left\{ e^{i\theta} \frac{\alpha - z}{1 - \overline{\alpha} z} \right\}.}$$

Similarly,

$$\boxed{\mathrm{Aut}(\mathbb{H}) \cong PSL_2(\mathbb{R}).}$$

Finally, the Riemann mapping theorem provides the global conclusion

$$\boxed{\begin{array}{c}\Omega \subsetneq \mathbb{C} \text{ simply connected} \\ \Downarrow \\ \Omega \text{ is conformally equivalent to } \mathbb{D}.\end{array}}$$

Thus the final progression of the course is

$$\boxed{\text{analysis} \longrightarrow \text{topology} \longrightarrow \text{geometry.}}$$

Holomorphic functions began as functions satisfying a very restrictive notion of differentiability. Cauchy's theory revealed their analytic rigidity; residues converted local singularities into global information; the logarithm revealed the role of topology; and conformal mappings show that the same analytic structure governs the geometry of planar domains.

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Any errors, omissions, or inaccuracies in these notes are my own.